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FLIGHT PLANNING

ATC Flight Plan

DEFINITIONS

Flight Plan: Specified information provided to air traffic service units, relative to an intended flight or portion of a flight of an aircraft.

Repetitive Flight Plan (RPL): A flight plan related to a series of frequently recurring, regularly operated individual flights with identical basic features, submitted by an operator for retention and repetitive use by ATS units. If a plan if filed for at least 10 times, it can be used as a RPL.

Filed flight plan: The flight plan as filed with an ATS unit by the pilot or a designated representative, without any subsequent changes.

Current Flight Plan: The flight plan, including changes, if any, brought about by subsequent clearances.

Estimated elapsed time: The estimated time required to proceed from one significant point to another.

Estimated off-block time: The estimated time at which the aircraft will commence movement associated with departure.

Estimated time of arrival: For IFR flights, the time at which it is estimated that the aircraft will arrive over that designated point, defined by reference to navigational aids, from which it is intended that an instrument approach will be commenced, or, if no navigational aid is associated with the aerodrome, the time at which the aircraft will arrive over the aerodrome.
For VFR flights, the time at which it is estimated the aircraft will arrive over the aerodrome.

USE OF DCT (DIRECT)

• If a **departure airfield** is **located** or **connected** to the ATS route then the **coded designator** of that route will be the first entry. If it is **not on** or **connected to** the ATS route the first entry will be **DCT** followed by the **joining point**, followed by the **designator** of the ATS route.
• If the **destination airfield** is **not on** or **connected to** an ATS route the last entry will be **DCT.**
• Use DCT between coded designators not connected by ATS routes.

• Use DCT between a designated reporting point and a position, denoted by a latitude and longitude or a bearing or distance from a Nav-aid, which is outside the ATS route.

ATC 8 FLIGHT PLAN

PRIORITY «≡ FF → ADDRESSEE(S) EGTTZQZX EGLLZPZX EGLLBAWD
EINNZQZX «≡

FILING TIME → ORIGINATOR EIDWZPZX «≡

SPECIFIC IDENTIFICATION OF ADDRESSEE(S) AND/OR ORIGINATOR

3 MESSAGE TYPE «≡ (FPL — 7 AIRCRAFT IDENTIFICATION BAW805 — 8 FLIGHT RULES I — TYPE OF FLIGHT S «≡

9 NUMBER — TYPE OF AIRCRAFT B737 — WAKE TURBULENCE CAT. / M — 10 EQUIPMENT — S/C «≡

13 DEPARTURE AERODROME EIDW — TIME 1100 «≡

15 CRUISING SPEED — N0430 LEVEL F290 → ROUTE UR14 STU UG1 WOD DCT OCK «≡

16 DESTINATION AERODROME — EGLL — TOTAL EET HR. MIN 0050 → ALTN AERODROME EGBB → 2ND ALTN AERODROME «≡

18 OTHER INFORMATION
— REG/GBGJG
STS/HOSP
EET/EGTT0015) «≡

SUPPLEMENTARY INFORMATION (NOT TO BE TRANSMITTED IN FPL MESSAGES)

19 ENDURANCE HR. MIN — E/ 0230 → PERSONS ON BOARD P/ 103 → EMERGENCY RADIO R/ UHF U VHF V ELBA E

SURVIVAL EQUIPMENT → S / POLAR ~~P~~ DESERT ~~D~~ MARITIME M JUNGLE ~~J~~ → JACKETS J / LIGHT L FLUORES F UHF ~~U~~ VHF ~~V~~

DINGHIES → D / NUMBER 10 → CAPACITY 150 → COVER C → COLOUR YELLOW

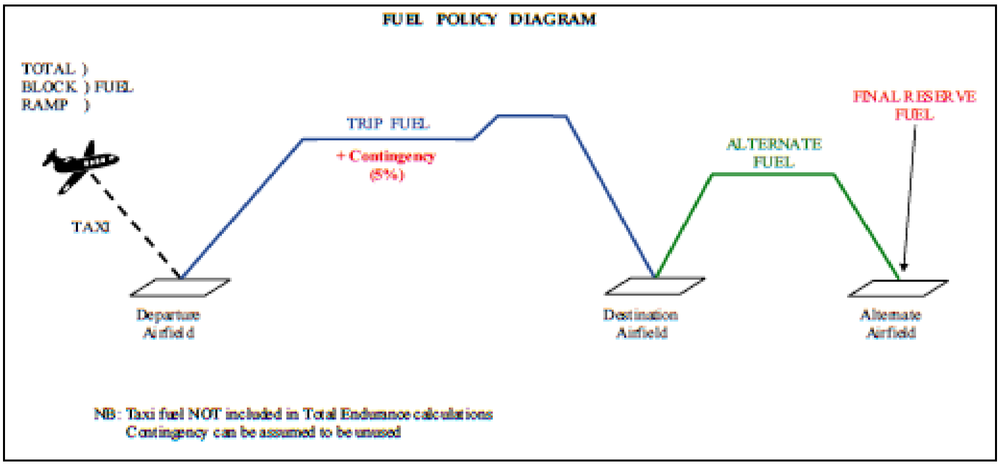
AIRCRAFT COLOUR AND MARKINGS A/ BLUE/GREY

REMARKS → N/ «≡

PILOT-IN-COMMAND C/ YENDLE) «≡

FILED BY BA OPS — SPACE RESERVED FOR ADDITIONAL REQUIREMENTS

Fuel Planning



Fuel is considered under the following breakdown:

1. **Taxi Fuel.** The amount required to start up, taxi, and hold (if necessary) before take-off. It will also include any fuel required to operate pre-flight services, such as cabin conditioning, APU etc.

2. **Trip Fuel.** This should include fuel:
   - For the take-off from the airfield elevation to the top of climb (TOC) at the initial cruising level/altitude.
   - From the TOC to top of descent (TOD), including any step climbs or descents.
   - From TOD to the point where the approach is initiated.
   - For approach and landing.

3. **Reserve Fuel.** Reserve fuel is further sub-divided into-
   - **I.** **Contingency Fuel**- A contingency is a chance occurrence or unforeseen event. Contingency Fuel is carried to compensate for deviations of aircraft from the bad weather conditions or any other planned routing.
   - **II.** **Alternate Fuel**- Fuel required to fly from missed approach at the destination to the planned alternate.
   - **III.** **Final Reserve**- a minimum landing fuel, and you should normally never land with less than the Final Reserve Fuel. It consists of 30 minutes (for a jet or turbo-prop) or 45 minutes (for a piston) fuel consumption at endurance speed.
   - **IV.** **Additional Fuel** – Fuel required when there is No En Route Alternate and Inability to Hold Height.
4. **Extra Fuel.** Extra fuel is any fuel above the minima required by Taxi, Trip and Reserve Fuel.

Calculation of Reserve Fuel –

- When ALTN is required (IFR Flight)-
  - Piston Engine - Fuel to DSTN + Fuel to ALTN + 45 min of Holding Fuel
  - Turbojet Engine- Fuel to DSTN + Fuel to ALTN + 30 min of Holding Fuel over ALTN @1500ft AGL + 5% of Contingency Fuel.
- When ALTN is not required (VFR Flight)-
  - Piston Engine- Fuel to DSTN + 45 Min of Holding Fuel
  - Turbojet Engine- Fuel to DSTN + 30 min of Holding Fuel @ 1500ft AGL.
- When ALTN is not available (Isolated Aerodrome) –
  - Piston Engine- 45 min of Fuel & 15% of Trip Fuel or 2 hrs of fuel consumption. (Whichever is less)
  - Turbojet Engine- Fuel to DSTN + 2 hrs of fuel at normal consumption.

Flight Planning Questions

1. **For a Repetitive flight plan to be used, flights must take place on a regular basis on at least-**
   a. 5 Occasions
   b. 20 Occasions
   c. 10 Occasions

2. **A current flight plan is**
   a. The flight plan as filed with an ATS unit by the pilot or a designated representative, without any subsequent changes.
   b. The flight plan, including changes, if any, brought about by subsequent clearances.
   c. The flight plan, including changes, if any, cleared prior to take off.
   d. The flight plan, including changes, if any, cleared prior to the aircraft's present position.

3. **ATC must be informed of changes which occur to the flight plan speed and ETA changes of ..... in TAS and .... of ETA be notified. Which answer fills the blanks correctly?**
   a. 3%, 5 minutes.
   b. 5 kts, 30 minutes.
   c. 5%, 3 minutes.
   d. 3 kts, 3 minutes.

4. **A filed flight plan is?.......**
   a. The flight plan as filed with an ATS unit by the pilot or a designated representative, without any subsequent changes.

b. The flight plan, including changes, if any, brought about by subsequent clearances.
c. The flight plan, including changes, if any, cleared prior to take off.
d. The flight plan, including changes, if any, cleared prior to the aircraft's present position.

5. **Flight plans for flights affected by Air Traffic Flow Management (ATFM) rules, must be filed at least....... before EOBT**.
   a. 3 hours.
   b. 1 hour.
   c. 30 minutes.
   d. Never less than 10 minutes

6. **Normally, flight plans should be filed on the ground at least......before clearance to start up is requested. Exceptionally, when it is not possible to meet this requirement, operators should never give.......**
   a. 30 minutes, less than 60 minutes.
   b. 60 minutes, less than 30 minutes.
   c. 3 hours, cause such trouble again.
   d. 3 hours, 30 minutes.

Flight Planning Answers

1	2	3	4	5	6
C	B	C	A	A	B

Payload

Basic Empty Mass- Aeroplane's mass plus standard items such as unusable fuel & fluids, fire extinguishers, pyrotechnics, emergency oxygen equipment, electronic equipments etc.

Dry Operating Mass. The total mass of the aeroplane ready for a specific type of operation excluding all usable fuel and traffic load. This mass includes items such as:

(1) Crew and crew baggage;
(2) Catering and removable passenger service equipment; and
(3) Potable water and lavatory chemicals.

Maximum Zero Fuel Mass. The maximum permissible mass of an aeroplane with no usable fuel.

Maximum Structural Landing Mass. The maximum permissible total aeroplane mass upon landing under normal circumstances.

Maximum Structural Take Off Mass. The maximum permissible total aeroplane mass at the start of the take-off run. The allowable Take-off Mass is the lowest of:

- **Regulated Take-off Mass**
- **Regulated Landing Mass + Fuel used in flight**
- **MZFM + Take-off Fuel**

Traffic Load. The total mass of passengers, baggage and cargo, including any nonrevenue load. The allowed traffic load is the lowest of the following

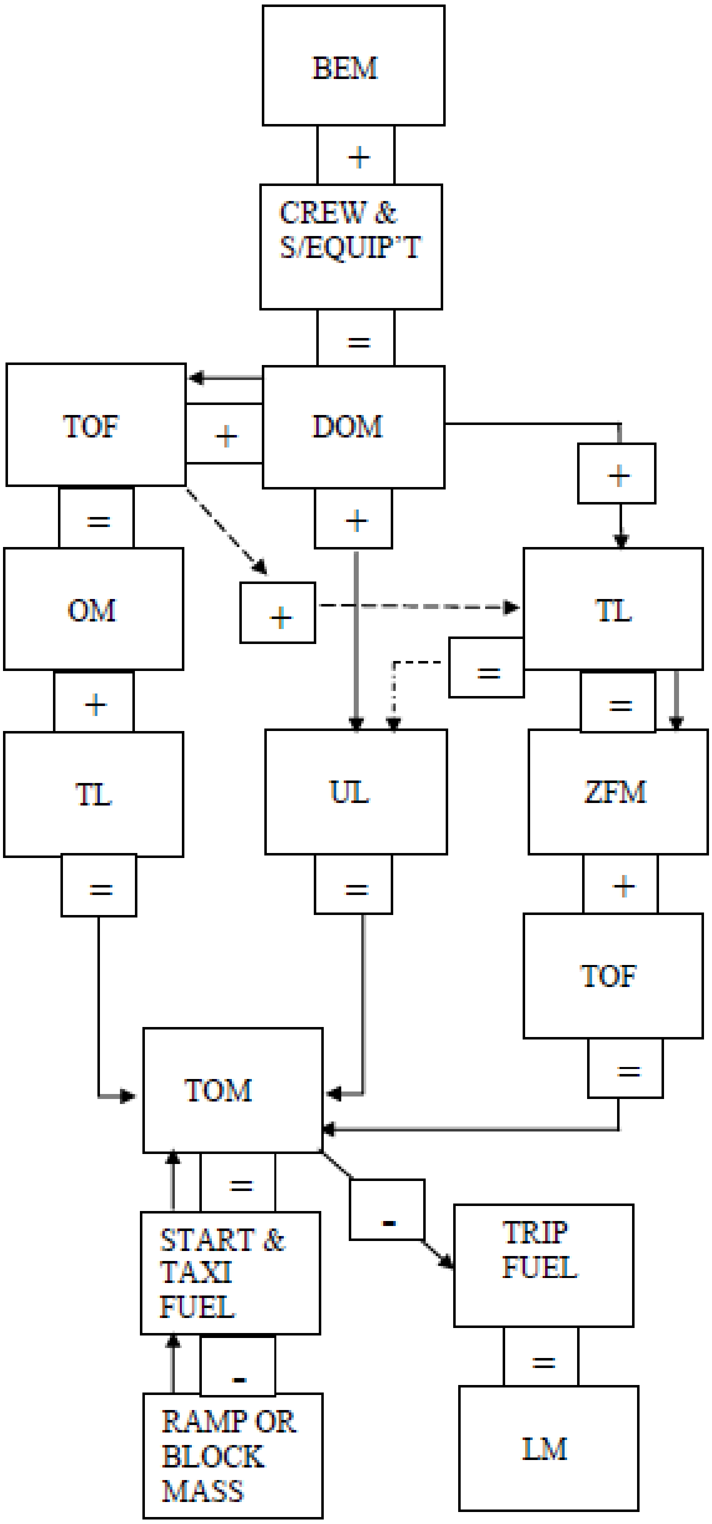
- **Structural Limited Traffic Load** = MZFM – DOM
- **Take-off Limited Traffic Load** = RTOM – DOM – Take-off fuel
- **Landing Limited Traffic Load** = RLM – DOM – Fuel remaining

Performance Limited Takeoff Mass – Take-off Mass limited due to performance factors such as wet runway, high temperature.

Regulated Takeoff Mass – It is the MTOM for a particular day calculated by temperature & other conditions. Lowest of Take-off Mass & Performance Limited Take-off Mass.
Maximum Ramp Mass –The maximum mass approved for ground manoeuvers. (MTOM + Taxi Fuel)
Useful Load - The Useful Load is the sum of the traffic load and the take-off fuel load.

Fuel On Board - Taxi Fuel + Trip Fuel + ALTN Fuel + Contingency Fuel (5% of Trip Fuel) + Final Reserve Fuel (30 min @ALTN @ 1500ft AGL @ Best Endurance Speed @ ISA Conditions & Nil Winds) + Extra Fuel (Arrival Delay).

Payload Summary Diagram



Payload Theory Questions

1. **The Dry Operating Mass is the total mass of the aeroplane ready for a specific type of operation and includes:**
   a. Crew and passenger baggage, special equipment, water and chemicals.
   b. Crew and their hold baggage, special equipment, water and contingency fuel.
   c. Crew baggage, catering and other special equipment, potage water and lavatory chemicals.
   d. Crew and baggage, catering and passenger service equipment, potable water and lavatory chemicals.

2. **The Maximum Structural Take-off Mass is:**
   a. The maximum permissible total aeroplane mass on completion of the refueling operation.
   b. The maximum permissible total aeroplane mass for take-off subject to the limiting conditions at the departure airfield.
   c. The maximum permissible total aeroplane mass for take-off but excluding fuel.
   d. The maximum permissible total aeroplane mass at the start of the take-off run.

3. **The Regulated Take-off Mass:**
   a. Is the lower of maximum TOM and the performance limited takeoff mass.
   b. Is the higher of the MZFM and the performance limited take-off mass.
   c. The maximum structural take-off mass subject to any last minute mass changes.
   d. The maximum performance limited take-off mass subject to any last minute mass changes.

4. **The Operating Mass:**
   a. Is the lower of the structural mass and the performance limited mass
   b. Is the higher of the structural mass and the performance limited mass
   c. Is the actual mass of the aircraft on take-off
   d. Is the dry operating mass and the fuel load.

5. **The Basic Empty Mass is the mass of the aeroplane:**
   a. Plus non-standard items such as lubricating oil, fire extinguishers, emergency oxygen equipment etc.
   b. Minus non-standard items such as lubricating oil, fire extinguishers, emergency oxygen equipment etc.
   c. Plus standard items such as unusable fluids, fire extinguishers, emergency oxygen equipment, supplementary electronics etc.
   d. Minus non-standard items such as unusable fluids, fire extinguishers, emergency oxygen and supplementary electronic equipment etc.

6. **The Traffic Load is:**
   a. The Zero Fuel Mass minus the DOM.
   b. The Take-off Mass minus the sum of the DOM and the total fuel load.
   c. The landing Mass minus the sum of the DOM and the mass of the remaining fuel.
   d. All the above

7. **The Basic Empty Mass is the:**
   a. MZFM minus both traffic load and the fuel load.
   b. Take-off mass minus the traffic load and the fuel load.
   c. Operating mass minus the crew and fuel load.
   d. Landing mass less traffic load.

8. **The term 'baggage' means:**
   a. Excess freight.
   b. Any non-human, non-animal cargo.
   c. Any frieght or cargo not carried on the person.
   d. Personal belongings.

9. **The Traffic Load:**
   a. Includes passenger masses and baggage masses but excludes any non-revenue load.
   b. Includes passenger masses, baggage masses and cargo masses but excludes any non- revenue load.
   c. Includes passenger masses, baggage masses, cargo masses and any non-revenue load.
   d. Includes passenger masses, baggage masses and any non-revenue load but excludes cargo.

10. **What is the maximum mass an airplane can be loaded to airplane before it moves under its own power?**
    a. Maximum Structural Ramp mass
    b. Maximum Structural take-off mass
    c. Maximum Regulated Ramp Mass
    d. Maximum Regulated Take-off mass

11. **Define the useful load:**
    a. traffic load plus dry operating mass
    b. traffic load plus usable fuel mass
    c. dry operating mass plus usable fuel load
    d. that part of the traffic load which generates revenue

Payload Numerical Questions

1. **Given –**
   MTOW = 45000kg
   APS weight = 22000kg
   MLW= 35,000kg
   MZFW= 33,000kg
   F/C = 1500 kg/hr
   FLT Time = 3hr 30min
   Reserve Fuel = 45 min.
   Calculate Max Payload.

2. **Given - MTOW = 48000kg**
   APS weight = 24000kg
   MLW= 37,000kg
   MZFW= 36,000kg
   FLT Time = 4hr 15min
   F/C = 1600kg/hr
   Reserve Fuel = 10% of Flight Fuel
   Fuel for taxi= 400 kg
   Fuel for ALT = 30min.
   Calculate Max Payload.

3. **Given –**
   RTOW = 50T
   MZFW = 39.1T
   MLW = 43.25T
   Trip Fuel = 4.2T
   Reserve Fuel = 1T
   APS = 27.5T
   Calculate Payload & Extra fuel that can be carried with the same payload.

4. **TOM 61,000 kg , LM 54,000 KG, MZFM 36,000 kg, Operating mass 55,000 kg, Trip fuel 30,000 kg, Contingency fuel 5% of trip fuel, Alternative fuel 500 kg, Final reserve 500 kg, Flight duration 3 hours, Fuel consumption 500 kg per hour per engine, Useful load 41500 kg**
   a. 58,000 kg
   b. 61,000 kg
   c. 56,145 kg
   d. 56,545 kg

5. **Given –**
   BOW = 27,000kg
   Trip Fuel = 3000kg
   Total Fuel = 7000kg
   MTOW = 49000kg
   MLW = 45000kg
   Max operating weight without fuel = 40,000kg.
   Calculate Max Payload.

6. **MTOW = 55,000kg , MLW = 42,500kg, MZFW = 38,000kg, Weight less Fuel & Payload = 29,000kg, Distance A to B = 1550nm, F/C = 1600kg/hr, Avg. TAS = 260kts, Reserve Fuel unused = 2750 kg. Calculate Max Payload in –**
   a. **Nil Winds**
   b. **90kts HW**

Payload Theory Answers

1	2	3	4	5	6	7	8	9	10	11
D	D	A	D	C	D	C	D	C	A	B

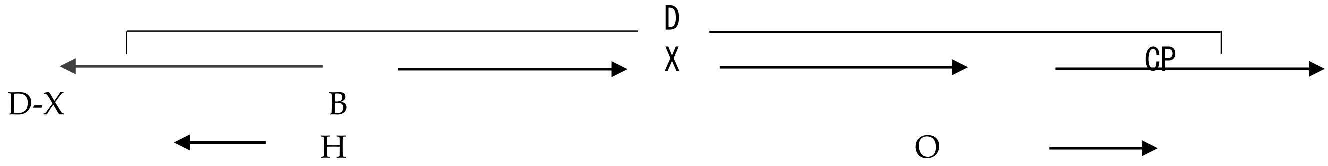
Payload Numerical Answers

1. 11,000kg
2. 11,520kg
3. 11.6T & 3.15T
4. A
5. 13000kg
6. 9000kg/ 8660kg

Critical Point & Point of No Return

Critical Point/ Point of Equal Return-

- Critical Point (CP) is the position from which A/c will take **the same time** to fly to either of two selected base.
- Required in the event of emergency in an aircraft and a landing is mandatory.
- Derivation of Formula-



By definition, O = H & Time = Distance/Groundspeed

$$\frac{X}{H} = \frac{D-X}{O}$$

$$XO = H(D-X)$$

$$XO = HD - HX$$

$$XO + XH = HD$$

$$X(O+H) = HD$$

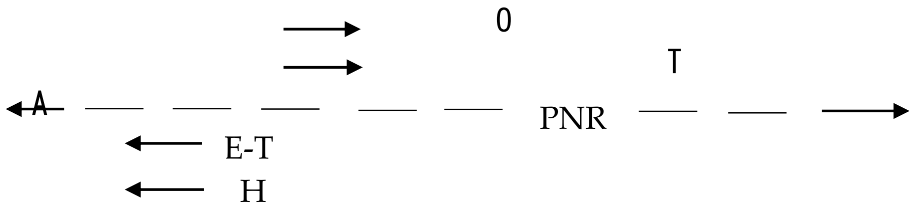
(Distance to CP) $\boxed{X = \frac{DH}{O+H}}$

$\boxed{\text{Time to CP} = \frac{X}{O} \text{ or } \frac{DH}{O(O+H)}}$

- In Nil Winds or Beam Winds the CP is HALFWAY.
- If there is a wind then the CP moves into the wind.
- The stronger the wind the greater the movement of CP into the wind **i**.e- In strong HW, CP moves towards destination & in strong TW, CP moves towards the departure.
- If TAS is increased, it has same effect as decrease in wind & CP moves towards departure field. Similarly, if TAS is reduced, CP moves towards destination field.
- CP is independent of fuel/endurance.
- In case of Beam Winds, DCP will remain same but TCP will increase as there will be little headwind to crab into the wind.

Point of Safe Return/ Point of No Return-

- **PNR** is the furthest point along a planned route to which an aircraft can fly and return to the departure airfield, or departure alternate, within the **SAFE ENDURANCE** of the aircraft.
- **Safe Endurance =** $\frac{\text{Fuel on board- Reserve Fuel}}{\text{Fuel Consumption/hr}}$
- Required when both destination & destination alternate are not available.
- Derivation of Formula-



By definition, Distance out to PNR = Distance home from PNR & Distance = Speed X Time,

$$T \times O = (E-T)\,H$$
$$TO = EH - TH$$
$$TO + TH = EH$$
$$T\,(O + H) = EH$$

(Time to PNR) $\boxed{T = \frac{EH}{O+H}}$

$\boxed{\text{Distance to PNR} = T \times O \text{ or } \frac{EOH}{O+H}}$

- Greatest DPNR occurs in still air.
- Change in wind direction does not affect DPNR.
- DPNR decreases in winds (Both HW & TW) or PNR moves towards departure point in winds.
- Abeam winds or Constant wind also reduces DPNR as compared to still conditions.
- DPNR varies directly with Endurance. i.e- FOB$\uparrow$, DPNR $\uparrow$.
- If A/c performance increases, DPNR also increases.
- If FOB is only Trip fuel + Reserve fuel (Safe Endurance = Flight Time), CP & PNR will coincide.
- PNR will usually be beyond CP.
- Still Air Range
  - Is the max. distance an A/c can fly in still air, and
  - Is equal to Total Endurance X TAS.

CP & PNR Questions

1. **Given- Distance from A to B 1200 nm, GS On 230 kt, GS Home 170 kt. What is the distance and time to the PET from "A"?**
   a. 600 nm 2 hr 37 min
   b. 510 nm 2 hr 13 min
   c. 690 nm 3 hr
   d. 510 nm 3 hr 33min

2. **Given- 2 Engine TAS 480 kt,1 Engine TAS 400 kt, W/V 330°/80, A to B 3500 nm, Course 200°. What is the distance and time to the engine failure PET from "A"?**
   a. 1515 nm 3 hr 23 min
   b. 1558 nm 2 hr 56 min
   c. 1515 nm 2 hr 51 min
   d. 1985 nm 3 hr 44 min

3. **CAS 190 kt cruising, Pressure altitude 9000 ft, Temperature ISA -10°C, W/V 320/40 kt, A to B is a distance 350 nm. Course 350°, Endurance 3 hours. What is the distance to the PET? And, given an actual time of departure (ATD) of 11:05, what is the ETA for the PET?**
   a. 220 nm & 12:49
   b. 311 nm & 12:26
   c. 146 nm & 11:55
   d. 204 nm & 12:13

4. **Flying from A to B, 270 nm, True track 030°, W/V 120°/35, TAS 125 kt. What are the distance and time to the Point of Equal Time?**
   a. 141 nm 65 mins
   b. 141 nm 68 mins
   c. 135 nm 68 mins
   d. 150 nm 65 mins

5. **When you have Beam Winds-**
   a. TAS= GS
   b. TAS < G/S
   c. TAS> G/S

6. **The CP is exactly half when-**
   a. O= H
   b. In nil winds conditions
   c. In beam winds conditions
   d. All are correct

7. Given: Total endurance 7hr 40 min, Safe endurance 6 hr, GS Out 230 kt, GS Home 170 kt. What is the time and distance to the PSR from "A"?

a. 2 hr 33 min 587 nm
b. 3 hr 15 min 750 nm
c. 3 hr 27 min 794 nm
d. 2 hr 33 min 434 nm

8. Given: TAS 500 kt, W/V 330°/50, A to B 4600 nm, Course 090°, Total endurance 12hrs, Safe endurance 10 hrs. What is the time and distance to the PSR from "A"?

a. 4 hr 45 min 2480 nm
b. 2 hr 22 min 1235 nm
c. 5 hr 42 min 2974 nm
d. 4 hr 45 min 2242 nm

9. Given: TAS 160 kt, W/V 100°/30, A to B 1620 nm, Course 030°, Depart A at 09:30 UTC, Total endurance 4hrs, Safe endurance 3hrs 20 min. What are the distance, time and estimate to the PSR from "A"?

a. 94min 231 nm 11:04
b. 106 min 261 nm 11:16
c. 128 min 315 nm 11:38
d. 106 min 296 nm 11:16

10. Given: GS Out 400 kt, Fuel flow out 2800 kg/hr, GS Home 450 kt, Fuel flow home 2500 kg/hr, Total endurance 15000 kg Reserves required 3000 kg. What is the distance and time to the PSR from "A"?

a. 1194 nm 3 hr
b. 872 nm 2 hr 11 min
c. 955 nm 2 hr 23 min
d. 1468 nm 3 hr 40 min

11. Given: 15,000 kg total fuel, Reserve 1,500 kg, TAS 440 kt, Wind component 45 head outbound, Average fuel flow 2150 kg/hr. What is the distance to the point of safe return?

a. 1520 nm
b. 1368 nm
c. 1702 nm
d. 1250 nm

12. The CP & PNR are collocated when-

a. O = H
b. In HW conditions
c. In Tail wind conditions
d. FOB is just sufficient for flight

13. The effect of an increasing strength of wind at 90^0 to track on DCP & TCP will be:-

Distance	**Time**
a. none	increase
b. decrease	decrease
c. none	none

14. The effect of position of CP with reducing TAS if there is a HW component is:-

a. increases the distance
b. decreases the distance
c. distance remains unchanged

15. The DPNR is

a. maximum in nil winds conditions
b. Increases with increase in TW
c. Decreases with decrease in TW
d. Is not affected by wind velocity.

16. Given: TAS 165 kt, W/V 090°/35, A to B 1620 nm, Course 035°. What is the distance and time to the CP from "A"

a. 903 nm 6 hr 04 min
b. 810 nm 5 hr 42 min
c. 708 nm 5 hr
d. 912 nm 6 hr 26 min

17. With a fuel of 10000lbs, the PNR is calculated to be 880nm. Other factor remaining constant if fuel is increased to 11000lbs, the distance to PNR is?

a. 928 nm
b. 968 nm
c. 950 nm
d. 920 nm

18. On a flight an aircraft is found to be achieving a G/S 10% higher than the planned G/S assuming all other conditions to be same the revised distance to PNR will be?

a. Less
b. More
c. No change
d. Impossible to say

19. The distance to PNR with 50 knots headwind is 1200nm. The distance in PNR with 50 knots tailwind will be?

a. Less than 1200nm
b. More than 1200nm
c. 1200nm
d. In could be any value

CP & PNR Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
D	C	D	C	C	D	A	B	B	C	B	D	A	A	A	D	B	A	C

CP & PNR Numericals

1. **Distance A to B is 925nm, track 153, TAS 210 knots, W/V 200/30, FOB 1000 GAL, Reserve 200 GAL, F/C 180 GPH. Find –**
   a. TCP and DCP
   b. TPNR and DPNR
   c. SAR

2. **On Flight from X to Y, Track 250, TAS 195 knots, Distance 975nm W/V 330/35. FOB 900 Gallons. F/C 150 GPH. Find –**
   a. TPNR & DPNR
   b. TCP & DCP
   c. SAR

3. **On a flight from A to B, Track 155, TAS 200 knots, W/V 260/35, DPNR 1050 nm, F/C 141 GPH. Find –**
   a. Fuel on board
   b. SAR
   c. If CP lies one hour 15 min before PNR find excess fuel carried on board.

4. **Distance A to B = 600 nm, Track = 080(T), W/V = 020/30, TAS = 250 kt. Calculate DCP.**

5. **DCP is 600nm with 30 kt H/W and total distance is 1000nm. If H/W changes to 30kt T/W DCP will be?**

6. **Aircraft departs point A on track 135 at 200 knots TAS for place B 600nm away. W/V=080/35. Calculate DCP and TCP in the case of engine failure with reduced TAS of 180 knots.**

7. **An a/c flies from A to B on track 150 and distance 1000nm with TAS 330 knots. W/V upto CP is 190/10 and 220/35 thereafter. Calculate DCP and TCP.**

8. **Distance A to B is 500nm, Track 060(T), W/V = 300/30, TAS = 250 kts, Safe endurance = 3hrs. Calculate DPNR.**

9. **The distance to PNR, 500nm is calculated with 5 hrs endurance and 30 knot H/W before the flight, the pilot experiences unexpected 30 knots T/W during flight, the new DPNR will be?**

10. **FOB = 6500 kg, FC = 500 kg/hr, Reserve fuel = 45 min, TAS = 250kt, Track = 145 (T), W/V = 300/30. Find DPNR and TPNR.**

11. **An A/C FC is 2000gph on 4 engines with 220 kts TAS and 1800gph on 3 engines and the TAS is 200knots with 20kt H/W on outbound. Calculate FOB with 1000 gallons reserve if DPNR is 600nm.**

12. **DPNR is 720nm with 10000lbs FOB. Calculate PNR with 10500lb fuel.**

13. **FOB = 2200lbs including 10% of flight fuel as reserve. Fuel flow normal 500lbs/hr and 450 lbs/hr on reduced power. TAS normal=240 knots reduced TAS=210 kts. Winds 10kt HW on outbound. Calculate DPNR and TPNR.**

14. **In case a/c is required to return to departure point using only 10000lbs fuel and flight fuel required is 15000lbs. FC = 5000lb, TAS = 400kt with HW of 100knots. The PNR and CP relationship is –**
    a. PNR is 188nm before CP
    b. PNR is 188nm from CP
    c. Both are collocated

15. **Track = 073, Distance = 1500nm, TAS = 210 kts, W/V = 200/35 for first 900nm then outwards 230/40, FOB = 1550GAL including reserve of 150GAL. FC = 200gph. Find-**
    a. SAR
    b. DCP/TCP
    c. DPNR and TPNR

16. **Track = 335, TAS = 195kt, Distance = 1275nm, W/V =260/35, FC = 105gph. Find –**
    a. DCP
    b. If TCP is 55 min before PNR. What is FOB excluding reserve.

CP & PNR Numerical Answers

1. a. 2:41 hrs, 507 nm
   b. 2:26 hrs 459 nm
   c. 1166nm

2. a. 2:42 hrs, 503 nm
   b. 2:45 hrs, 511 nm
   c. 1170 nm

3. 10:22 hrs
   b. 1338 nm
   c. 1652 IST

4. 318nm

5. 400nm

6. 334nm, 1hr52min

7. 516, 1hr36min

8. 561, 3hr31min

9. 372nm, 750nm

10. 500nm

11. 1512nm, 5hrs27min

12. 11909

13. 756nm

14. A

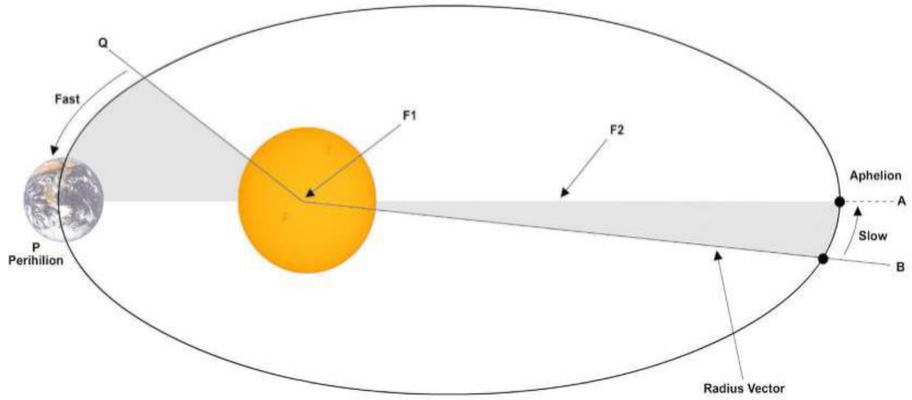
15. a. 1628nm
    b. 656nm, 2hr52min
    c. 721nm, 3hr9min

16. DCP 667nm, 917GAL

Solar System

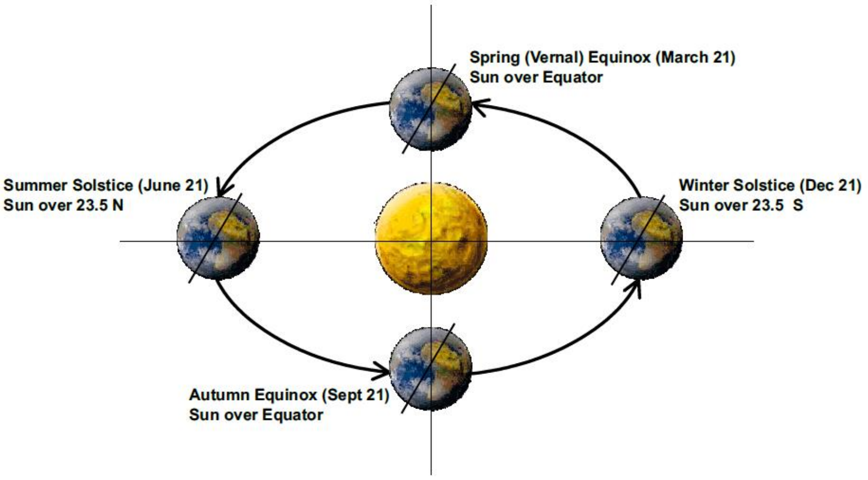
Kepler's Laws of Planetary Motions

- The orbit of each planet is an ellipse with the sun at one of the foci.
- The line joining the planet to the sun, known as the radius vector, sweeps out equal areas in equal time.



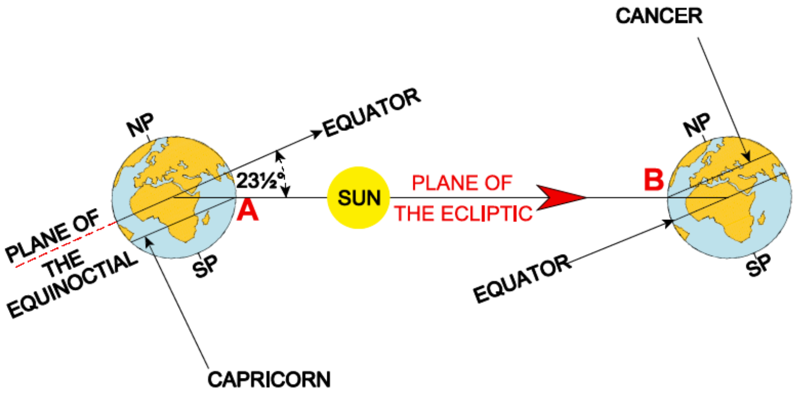
THE SEASONS

- The predominant cause of the Seasons is the inclination (tilt) of the earth.
- The Earth's axis is inclined (tilted) at an angle of 66.5° to its orbital plane and this is often stated as 23.5° to the normal to the orbital plane.
- The point where planet is closest to the Sun is known as Perihelion (early January) & the point where sun is farthest is known as Aphelion (early July).



- The sun appears vertically above 23½°S on the 21st of December. This is known as Winter Solstice in NH & Summer Solstice in SH.
- The sun will appear above 23½°N on the 21st of June. This is known as Summer Solstice in NH & Winter Solstice in SH.
- The sun crosses the equator from South to North on about 21st March. This is known as Spring Equinox in NH & Autumn Equinox in SH.

- Six months later it crosses the equator from North to South on about 21st September- NH - Autumn Equinox SH - Spring Equinox.



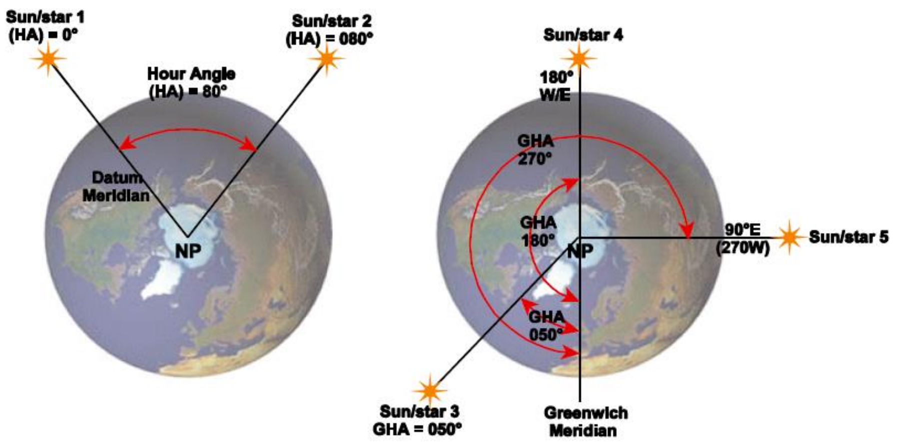
- The plane of the equator is called the 'Plane of the Equinoctial'. (Equal day/night).
- The Plane of the Ecliptic and the Plane of the Equator are inclined to each other at an angle of 23.5°.
- At a given time of year, the angle that the Sun is above or below the equator determines the season and affects the length of daylight/night. The angle is known as DECLINATION.
- The Sun's declination changes annually between 23.5°N (Sun overhead the tropic of Cancer) to 23.5°S (Sun overhead the Tropic of Capricorn) and then back to 23.5°N.
- The rate of change of the length of daylight will therefore be greatest when the rate of change of declination is greatest. This situation occurs at the equinoxes (about Mar 21 and Sep 21).

Measurement of Days & Years

- Day & night are formed since earth rotated on its axis from west to east direction.
- A 'day' may be defined as the length of time taken for the Earth to rotate once about its axis measured against a celestial body (the Sun or a star). Measurements against a star are called 'sidereal' and against the Sun are called 'solar'.
- **Sidereal Day**- Is measured against a distant star and is of nearly constant length.
- **Apparent Solar Day**- It is measured against the Real or Apparent Sun. However, due to Kepler's second law, speed of earth around the ecliptic varies as the distance from the sun varies. Apparent Solar Day will not be a constant value (23hrs 44 min to 24hrs 14 min).
- The maximum difference between Mean Time and apparent (real) sun time is about 16 minutes and occurs in Nov & Feb. The difference is known as the **Equation of Time.**
- An apparent solar day is longer than a sidereal day.
- **Mean Solar Day**- Is the average length of an apparent solar day and has an exact value of 24 hrs. It can be smaller or longer than apparent solar day.
- **A Sidereal Year** is the time taken by the Earth to complete an orbit of the Sun measured against a distant star. Its length is 365 days 6hrs 9 min.

- **An Apparent Solar Year** (Tropical Year) is the length of one cycle of the seasons. Its length is 365 days, 5 hrs and 49 minutes (20 min shorter than sidereal year).
- **A Calendar Year** is normally 365 days. It is kept in step with the tropical year by adding a day every 4th year, a 'leap' year. A fine adjustment is made on 3 occasions every 400 years.

Hour Angle

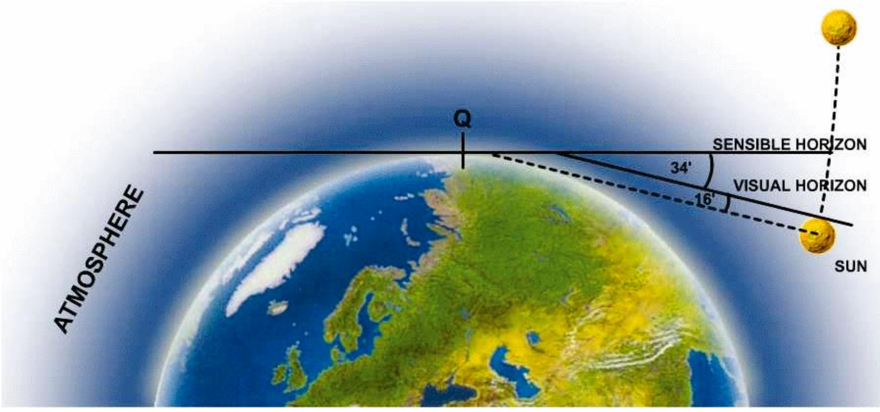


- The Earth spins in an easterly direction, 360° in every 24 hours. Thus, a celestial body (Sun) will transit across a given meridian at 24 hour intervals.
- The Hour Angle of a celestial body is defined as the arc of the Equator (equinoctial) intercepted between the meridian of a datum and the meridian of the body, measured westwards from 0° to 360°.
- Thus when a celestial body transits a given meridian, its Hour Angle is 000°. When the body transits the anti-meridian, its Hour Angle is 180°.
- If the given meridian is Greenwich, the Hour Angle is known as the Greenwich Hour Angle (GHA) which is directly analogous to longitude. A body with a GHA of 040° will be transiting the 040W meridian. But a body with a GHA of 270° will be transiting the 270W meridian, or the 090E meridian.

Twilight

- Twilight is that period before sunrise and after sunset when refracted light from the Earth's atmosphere gives an amount of illumination.
- Twilight occurs as due to atmospheric refraction, the 'visual' horizon is below the 'sensible' horizon.
- **Civil twilight** occurs when the Sun's center is between 0° and 6° below the sensible horizon.
- **Nautical Twilight**- The center of the Sun is between 6° and 12° below the sensible horizon.
- **Astronomical Twilight**- The center of the Sun is between 12° and 18° below the sensible horizon (complete darkness).
- At Equator, twilight period is 21 minutes.

- With increase in latitude, the twilight period increases.



Solar System Questions

1. **When does perihelion occur?**
   a. early January
   b. mid March
   c. early July
   d. September 21

2. **Viewed from the North Celestial Pole (above the North Pole), the Earth orbits the Sun:**
   a. clockwise in a circular orbit
   b. anti-clockwise in a circular orbit
   c. clockwise in an elliptical orbit
   d. anti-clockwise in an elliptical orbit

3. **When do 'equinoxes' occur?**
   a. December and June
   b. February and November
   c. March and September
   d. January and July

4. **When it is the Winter Solstice in the Southern Hemisphere, the Declination of the Sun is:**
   a. 0°N/S
   b. 23½°N
   c. 66½°N
   d. 23½°S

5. **What is the angle between the Equinoctial and the Ecliptic?**
   a. 66½°
   b. 23½°
   c. varies between 23½°N and 23½°S

d. varies between 66½°N and 66½°S

6. **When is the rate of change of the length of daylight greatest?**
   a. February/November
   b. January/July
   c. at the Equinoxes
   d. at the Solstices

7. **A sidereal day is?**
   a. longer than an apparent solar day
   b. longer than a real solar day
   c. shorter than an apparent solar day
   d. equal to a real solar day

8. **The maximum difference between mean noon (1200LMT) and real/apparent noon occurs in?**
   a. January/July
   b. March/September
   c. November/February
   d. December/June

9. **The maximum difference between Mean Time and Apparent Time is:**
   a. 21 minutes
   b. 16 minutes
   c. 30 minutes
   d. there is no difference

10. **What is the length of a Sidereal Year?**
   a. 365 days
   b. 366 days
   c. 365 days 6 hrs
   d. 365 days 5 hrs 49 minutes

11. **'The Calendar Year and the Tropical Year are of different lengths. The difference is adjusted partly by using leap years every fourth calendar year. However, some years are not designated as leap years'. Which of the following years will be a leap year?**
   a. 2001
   b. 2100
   c. 2300
   d. 2400

12. **The Hour Angle (Greenwich Hour Angle) of a celestial body is analogous/equivalent on the Earth to ________?**
   a. latitude

b. longitude
c. co-latitude
d. UTC

13. A star has a Greenwich Hour Angle (GHA) of 220°. Which meridian is the star transiting?
a. 040W
b. 040E
c. 140W
d. 140E

14. Nautical Twilight occurs when the sun is between _________ and ____________ below the Sensible Horizon?
a. 0°/6°
b. 6°/12°
c. 12°/18°
d. 18°/24°

15. Between 60N and 60S, the minimum duration of Civil Twilight is?
a. 21 minutes
b. 16 minutes
c. 14 minutes
d. 30 minutes

16. On Mid-summer Day in the Southern Hemisphere, the sun will be overhead:
a. 66½ S
b. 23½ N
c. 23½ S
d. the Equator

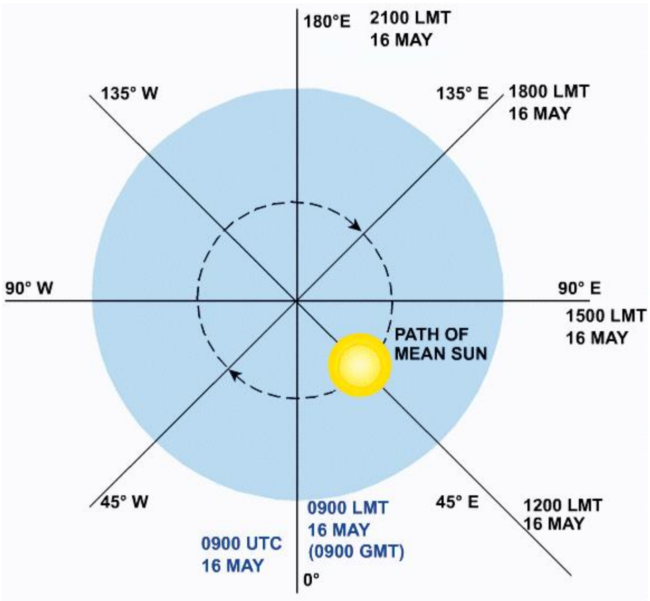
17. On Mid-winter Day in the Northern Hemisphere, the sun will be overhead:
a. 66½ S
b. 23½ N
c. 23½ S
d. the Equator

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
A	D	C	B	B	C	C	C	B	C	D	B	D	B	A	C	C

Time

Local Mean Time-



- Time kept at a meridian with respect to position of sun at observer's anti-meridian is called Local Mean Time (LMT).
- All places on same meridian will have same LMT & it will change on change of meridians.
- LMT uses local meridian of longitude as its reference point.
- Further east to the place, further ahead is LMT.

Greenwich Mean Time-

- GMT is LMT at Greenwich Meridian.
- Also called UTC, Coordinated Universal Time. UTC & GMT are same for all practical purposes.
- It remains same all over the world at a particular time.
- Long East- GMT Least, Long West- GMT Best.

Conversion of Arc (Angle) to Time-

360° = 24 hrs
15° = 1 hour
1° = 4 minutes
15′ of a degree of arc = 1 minute of time
15″(seconds of arc) = 1 second of time

Longitude	LMT	GMT/ UTC	Longitude	LMT
$75^0$00′W	030150		$30^0$00′E	
$62^0$15′E	172135		$146^0$30′E	
$147^0$23′W	230046		$168^0$56′W	
$84^0$37′W	282216		$12^0$52′W	
$159^0$19′W		280902	$110^0$53′W	
$111^0$31′W		060446	$158^0$17′E	
$163^0$19′W		280902	$178^0$43′E	

Standard Time-

- It is the time set by legal authorities.
- Every longitude will have a different LMT.
- Each country maintains same time irrespective of change of longitudes and is called standard time.
- India maintains LMT at $82^0$30′E as standard time and is termed as INDIAN STANDARD TIME (IST).
- **IST = GMT + 5 hours 30 minutes**.

Zone Time-

- 360^0 longitudes is divided into 24 time zones with each time zone of 15^0 starting from A-Z except letters I and O.
- Zone Number is the value to be added to zone time algebraically to find UTC.

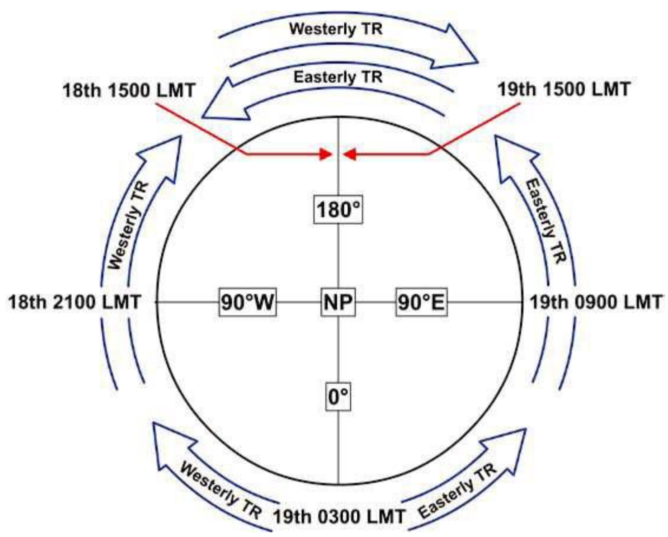
ZN= Longitude ÷ 15

When value is ≤ 0.5, Use lower time zone.

When value is > 0.5, Use higher zone time.

ZT = UTC ± ZN (East +, West -)

International Date Line-



EAST to West Day is Gained, Date Lost & LMT Decreases.

West to East Day is Lost, Date Gained & LMT Increases.

Time Questions

1. **Convert 153°30′ of arc of longitude into time.**
   a. 10 hrs 24 mins
   b. 10 hrs 22 mins
   c. 10 hrs 14 mins
   d. 10 hrs 8 mins

2. **The definition of Local Mean Time (LMT) is:**
   a. time based upon the average movement of the Earth around the Sun.
   b. when the Mean Sun is transiting (crossing) your meridian, it is 1200hrs LMT.
   c. when the Mean Sun is transiting (crossing) your anti-meridian, it is 0000hrs LMT (2400 hrs LMT, previous day).
   d. all of the above.

3. **Local Mean Time (LMT) always changes by a day when crossing __________?**
   a. the Greenwich Meridian
   b. 180°E/W
   c. the International Date Line
   d. the Equator

4. **When LMT at 4600N 00830W is 2300 on the 9th May, what is the LMT and local date at:**
   a. 4600N 10830W?
   b. 4600N 10830E?

5. **The time is 1300 UTC on 1st April. Calculate the LMT and local date at:**
   a. 5430N 00715E
   b. 1627S 10743W
   c. 1846N 16835E

Answers

1	2	3	4	5
C	D	B	A	B

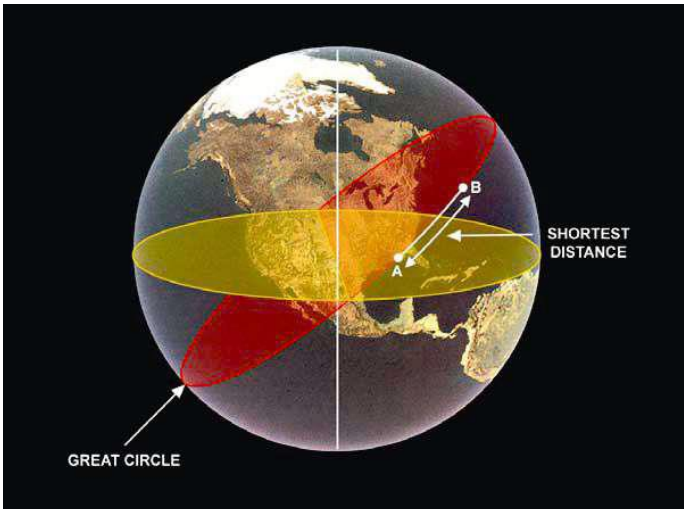
Earth

Basic Points-

- Shape of earth is Oblate Spheroid or Ellipsoid.
- Rotation of earth is anti-clockwise when viewed from North Pole & Clockwise when viewed from South Pole.
- Compression of Earth = $\frac{\text{Equatorial Diameter} - \text{Polar Diameter}}{\text{Equatorial Diameter}}$
- Value of compression is 1/297 ≅ 1/300 or 0.3%

Circles of Earth-

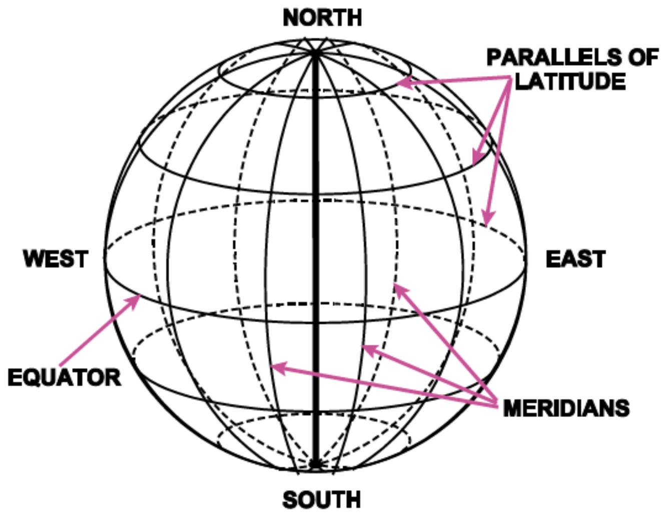
- **Great Circle**: A circle on the surface of the earth whose center and radius are those of the earth itself is called a Great Circle.



- **Equator**- The Great Circle whose plane is at 90° to the axis of rotation of the earth (the polar axis). It lies in an east-west direction & divides the earth into two hemispheres.
- **Meridians** are semi-Great circles joining the North and South poles. All meridians indicate True North-South direction. Every Great Circle passing through the poles forms a meridian and its Anti-meridian. The meridians cross the Equator at 90°.
- **Prime Meridian/ Greenwich Meridian-** Semi-Great Circle passing through Greenwich. It is the datum meridian for defining longitudes. It defines a longitude of 0^0 E/W & its anti-meridian lies at 180^0 E/W.
- A circle on the surface of the earth whose center and radius are not those of the earth and is smaller than Great Circle is called a **Small Circle.** Eg- Parallels of Latitude**.**

- **The parallels of latitude** are small circles on the surface of the earth whose planes are parallel to the Equator. They lie in an East-West direction. Their function is to indicate position North or South of the Equator.

- The network formed on a map or the surface of a globe by Prime Meridian, meridians, equator and the parallels of latitudes is called the **Graticule.**



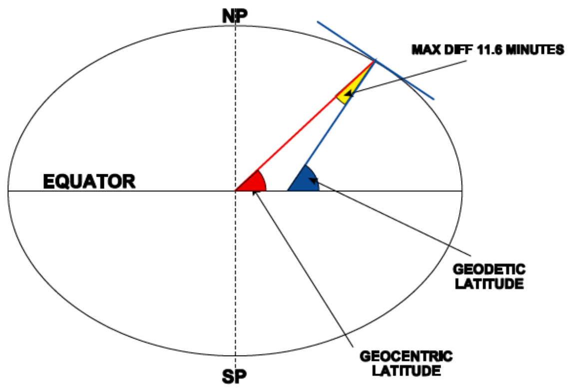
- **The Latitude** of any point is the arc measured along the meridian through the point from the Equator to the point. It is termed north/south depending upon position w.r.t. equator. Maximum latitude can be 90^0 N/S.

- **The longitude** of any point is the shorter distance in the arc along the Equator between the Prime Meridian and the meridian through the point. Longitude is measured in degrees and minutes of arc and is annotated E/W depending whether the point lies East or West of the Prime Meridian. Maximum longitude can be 180^0E/W.

Special Cases of Parallels of Latitude-

- **Arctic Circle** the parallel of 66½°N (note that 66½° is value of the Earth's tilt).
- **Antarctic Circle** the parallel of 66½°S
- **Tropic of Cancer** the parallel of 23½°N (the sun is overhead the Tropic of Cancer on mid-summers day in the NH)
- **Tropic of Capricorn** the parallel of 23½°S (the sun is overhead the Tropic of Capricorn on mid- winters day in the NH).

Geocentric and Geodetic Latitude-



- The angle between the line joining the point to the Geo-center of the Earth and the plane of the Equator. This is **Geocentric Latitude.**
- **Geodetic** (or Geographic) Latitude is the angle between the normal (90°) to the meridian at the point on the spheroid and the plane of the equator.
- The maximum difference between Geocentric and Geodetic Latitudes occurs at approx **45° N/S** and is about **11.6 minutes of arc.**
- The difference is zero at equator and poles.

Angular Measurements-

- A degree is sub-divided into 60 minutes of arc (‘)
- A minute can be further sub-divided into 60 seconds of arc (“).
- Note- They are not minutes or seconds of watch.

Change of Latitude (CH LAT) or Difference of Latitude (D LAT)-

- It is the difference in latitude of two places.
- Termed N/S depending on whether the change is in N or S direction.

Rule –

1. N to S = + Add
2. N to N = - Subtract
3. S to S = - Subtract

Change of Longitude (CH LONG) or Difference of Longitude (D LONG)-

- It is the difference in latitude of two places.
- Termed N/S depending on whether the change is in N or S direction.

Rule –

1. E to W = + Add
2. E to E = - Subtract
3. To go from backside of the earth, add the two longitudes & subtract from 360 & then check direction.

CH LAT / CHLONG Questions-

1. **What is the Change of Latitude between the following positions:-**
   a. 52°15′N to 39°35′N
   b. 49°35′N to 60°20′S
   c. 74°20′S to 34°30′S
   d. 71°20′N to 86°45′N over the north pole.

2. **What is the Change of Longitude between the following positions:-**
   a. 075°40′W to 125°35′W
   b. 001°20′E to 004°20′W
   c. 150°40′E to 179°30′E
   d. 162°36′W to 140°42′E

3. **Give the Direction and Change of Latitude and Longitude from X to Y in each case:**
   X Y
   a. 50°31′N 006°30′W to 52°00′N 008°35′W
   b. 47°32′N 002°46′W to 43°56′N 001°33′W
   c. 61°47′N 003°46′W to 62°13′N 001°36′E
   d. 31°27′S 091°47′E to 35°57′N 096°31′E

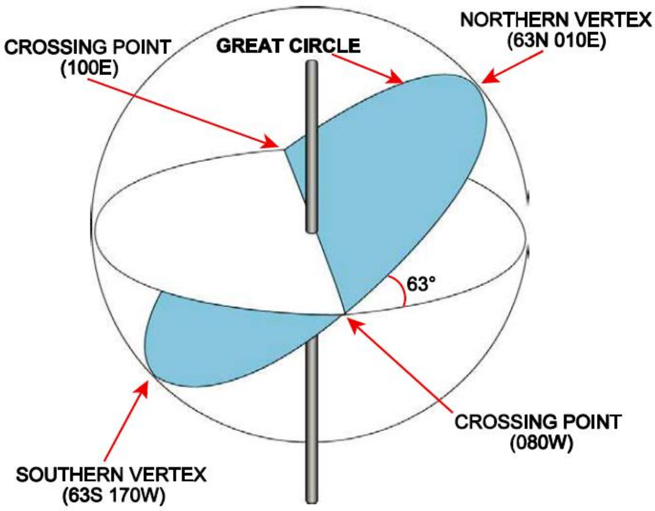
GREAT CIRCLE VERTICES-

Rules for Antipod –

1. For southern vortex, North Latitude become South Latitude. Eg- 63N becomes 63S.
2. For Longitude, add 180^0 to longitude & then subtract by 360^0. Eg- 10^0E + 180^0 = 190^0 - 360^0=170^0 W.

   OR

   Add 90^0 & subtract 90^0 from the longitude of equator where GC cuts the equator.
   Eg- 80^0W + 90^0 = 170^0 W & 80^0W - 90^0 = 10^0E.

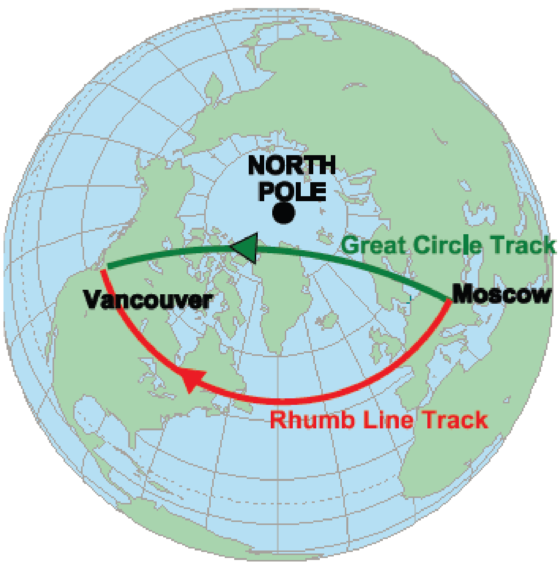


Question - A great circle has its North vertex at 70N 130E. What is the position of its South vertex?
At what longitudes and in what direction would the Great Circle cross the Equator assuming:

a. Initial direction is east from the Northern Vertex?
b. Initial direction is west from the Northern Vertex?

Properties of Great Circle –

- Largest possible circle that can be described on the surface of earth which divides earth in two equal halves.
- Has the same radius and center as the earth.
- On earth surface it appears as straight line.
- Given two points on the Earth's surface, there will be only one Great Circle joining them (unless the points are diametrically opposed).
- Shorter arc of Great Circle is the shortest distance between two points on earth.
- Radio Waves follow GC path. Eg- VOR, DME, ILS etc.



Properties of Rhumb Line –

- A Rhumb Line is a regularly curved line on the surface of the Earth which cuts all meridian at the same angle - **a line of constant direction**.
- Along a RL, A/c follows a constant direction.
- Any RL spirals up towards the nearest pole.
- RL distance is always more than GC distance.
- All Parallel of latitudes are RL.
- Equator is a GC as well as Rhumb Line.

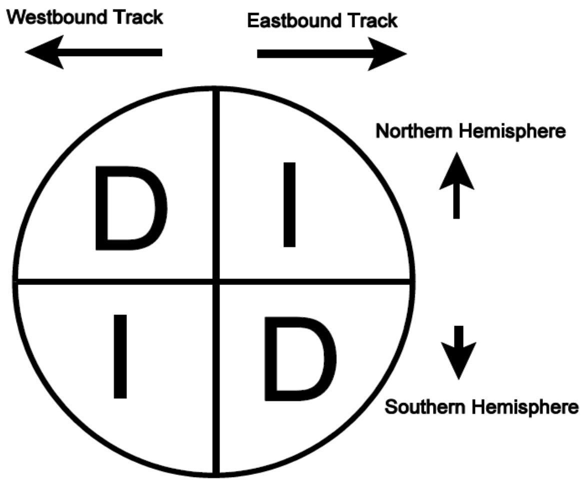
Note-

- The only lines which are both Great Circles and Rhumb lines are the Equator and any meridian (along with its associated anti-meridian).

- The parallels of latitude are Rhumb Lines because they cut all meridians at 90°, but they are Small Circles as they do not have the same radius and centre as the earth.

Great Circle Direction-

- **Northern Hemisphere**.
  - Consider an easterly track- a Rhumb Line track of 090°. If heading in an Easterly direction, the Great Circle track starts with an initial direction of about 030°, then curves round to 090° and finishes up on about 150°.Hence, the track direction is increasing.
  - Heading in a Westerly direction, the Great Circle track starts with an initial direction of about 330°, then curves round to 270° and finishes up on about 210°. Here, the track direction is decreasing.



- **Southern Hemisphere**.
  - Consider easterly track- a Rhumb Line track of 090°. If heading in an Easterly direction, the Great Circle track starts with an initial direction of about 150^0, then curves round to 090° and finishes up on about 030°. In other words, the track direction is decreasing.
  - If heading in a Westerly direction, the Great Circle track starts with an initial direction of about 210°, then curves round to 270° and finishes up on about 330°. In other words, the track direction is increasing.

Distance on the Earth-

1 metre (m) = 100 centimetres (cm) = 1000 millimetres (mm)
1 centimetre (cm) = 10 millimetres (mm)
1 metre (m) = 3.28 feet (ft)
1 foot (ft) = 12 inches ('in' or ")
1 inch (in) = 2.54 centimetres (cm)
1 yard (yd) = 3 feet (ft)

1 Kilometre (km = 3280 feet (ft)

- **The Kilometre** (km). The definition of the Kilometre is 1/10,000th of the average distance on the Earth between the Equator and either Pole.
- **The Statute Mile**- 1 SM= 5280 feet. It is hardly used in aviation.
- **Nautical Mile**-
  - The ICAO definition of the nautical mile is that it is a measure of distance of 1852 m.
  - The Standard Nautical Mile is defined as a length of 6080 feet.
  - Arc of GC/ Meridian which subtends an angle of 1 Minute at the center of the earth.
  - Length of Nautical Mile is more at Pole (1862m) than Equator (1843m).
  - Length of Nautical Mile is more at height.

One minute of latitude = 1 nautical mile (nm)

One degree of latitude = 60 minutes = 60 nm

But

One minute of Longitude = 1 nm AT THE EQUATOR ONLY.

Great Circle Distances

Case 1- What is the shortest (great circle) distance between A (5137N 00012W) and B (0648N 00012W)? (Same meridian, same hemisphere)

Case 2 - What is the shortest distance between D (2930S 03030E) and E (5947N 03030E)? (Same meridian, different hemispheres)

Case 3 - What is the shortest distance between F (4155N 01110E) and G (2117N 16850W)? (Meridian and anti-meridian, same hemisphere).

Case 4 - What is the shortest distance between J (3557N 13535E) and K (2210S 04425W)? (Meridian and anti-meridian, different hemispheres).

Case 5 - What is the shortest distance between L (0000N 01635W) and M (0000N 10355E)? (Two points on the Equator).

Case 6 - What is the shortest distance between N (5130N 00000E) and P (5130S 18000E)? (A special case)

Questions

1. **What is the approximate compression of the Earth?**
   a. 3%
   b. 0.03%
   c. 0.3%
   d. 1/3000

2. **A Graticule is the name given to:-**
   a. A series of lines drawn on a chart
   b. A series of Latitude and Longitude lines drawn on a chart or map
   c. A selection of small circles as you get nearer to either pole

3. **You are at position A at 54°20′N 002°30′W. Given a ch.lat of 16°20′N and a ch.long of 020°30′W to B, what is the position of B?**

4. **You are at position C at 36°47′S 179°21′E. Given a ch. lat of 46°47′N and a ch. long of 20°30′E to D, what is the position of D?**

5. **The diameter of earth is approximately:-**
   a. 6350 km
   b. 12700km
   c. 18500 km
   d. 40000km

6. **Any Meridian Line is a:**
   a. Rhumb Line
   b. Semi Great Circle
   c. Rhumb Line and a semi Great Circle

7. **The maximum difference between geocentric and geodetic latitude occurs at-**
   a. 45^0 N/S

b. 60^0 N/S
c. 90^0 N/S
d. Equator

8. An A/c at position 60N005W tracks 90^0(T) for 315km. On completion of flight the longitude will be –
a. $005^015'$E
b.$000^040'$E
c. $002^010'$W
d. $000^015'$E

9. An A/c at latitude 10^0N flies south at a groundspeed of 445 km/hr. What will be the latitude after 3 hr?
a. $12^015'$S
b.$22^000'$S
c. $03^050'$S
d. $02^000'$S

10. A Rhumb Line cuts all meridians at the same angle. This gives:
a. The shortest distance between two points.
b. A line which could never be a great circle track
c. A line of constant direction

Answers

1	C
2	B
3	70° 40′ N 023° 00′ W
4	10° 00′ N 160° 09′ W
5	B
6	C
7	A
8	B
9	D
10	C

Departure

- Equator is a GC where 1′ = 1 NM. As latitude increases, the spacing b/w the two meridians reduces and becomes zero at poles.
- That means it is maximum at 0^0 (equator) and minimum at 90^0 (poles). Therefore, this follows a cosine pattern.
- Departure is the **distance between two meridians** along a specified parallel of latitude.
- Departure is a RL distance. Hence, it is not the shortest distance.

Departure (nm) = CH LONG (deg) x 60 x Cos Lat

- If latitudes are different then, **Departure = CH LONG x 60 Cos Mean Lat.**
- If Departure at one latitude is given-

$$\frac{\textbf{Departure at Latitude A}}{\textbf{Departure at Latitude B}} = \frac{\textbf{Cos A}}{\textbf{Cos B}}$$

Questions

1. **A flight is to be made along the parallel of latitude from A at 48°00′N 04°00′W to B at 48°00′N02°27′E. Calculate the distance.**

2. **An aircraft flies for 1000nm along a RL track of 090°(T) from C at 36°00′N 174°45′E to D. What is the longitude of D?**

3. **The following rhumb line tracks and distances flown; starting from E in latitude 50°N.**
   E to F 000°(T) 300nm, F to G 090°(T) 300nm, G to H 180°(T) 300nm. What is the rhumb line bearing and distance of H from E?

4. **What is the track and distance measured along the parallel of latitude of 80°S from 176°15′W to 179°45′E?**

5. **In which latitude is a difference in longitude of 44°10′ the equivalent of a departure of 2295nm?**

6. **An aircraft departs a point 0400N 17000W and flies 600 nm South, followed by 600 nm East, then 600 nm North, then 600 nm West. What is its final position?**
   a. 0400N 17000W
   b. 0600S 17000W
   c. 0400N 169° 58.1′W
   d. 0400N 170° 018′W

7. An aircraft is flying around the Earth eastwards along the 60N parallel of latitude at a groundspeed of 240 knots. At what groundspeed would another aircraft have to fly eastwards along the Equator to fly once round the Earth in the same journey time?

a. 600 knots
b. 240 knots
c. 480 knots
d. 120 knots

8. Your position is 5833N 17400W. You fly exactly 6 nm eastwards. What is your new position?

a. 5833N 17411.5W
b. 5833N 17355W
c. 5833N 17340W
d. 5833N 17348.5W

Answers

1	259 nm
2	164°39′W
3	090° at 336 nm
4	270°(T) Dist 41.7 nm
5	30°N or S
6	C
7	C
8	D

SCALE

- Scale is the relationship between the length of a line drawn between two positions on a chart and the distance on the earth between the same points.
- It is **the ratio of map distance and earth distance** that it represents on earth's surface. (Both measured in same units).

$$\textbf{Scale} = \frac{\textbf{Chart Length}}{\textbf{Earth Surface}}$$

- **Representative Fraction**- It is the ratio of map difference with earth distance with numerator as 1.
- R.F = $\frac{1}{10,000}$ means 1 unit length on map equals 10,000 such unit items on earth.

Reminder about conversion factors-

1 nautical mile = 6080 feet or 1852 metres
1 metre = 3.28 feet
1 inch = 2.54 cm
5.4 nm = 10 kilometres

Questions

1. **Give the Scale, as a representative fraction, of the following charts:-**
   a. One Centimetre represents Five Kilometres
   b. One Centimetre represents 5.4 Nautical Miles
   c. One Inch represents 15.4 Kilometres
   d. 3.5 Inches represents 70 Kilometres
   e. Five Inches represents Eight Nautical Miles

2. **The Representative Fraction of a chart is given as 1: 500,000 (Half Mil Chart)**
   a. How many (1) Centimetres and (2) Inches would represent 30 Kilometres on the Ground?
   b. How many (1) Inches and (2) Centimetres would represent 30 Nautical Miles on the ground?

3. **The Representative Fraction of a chart is given as 1: 1000,000. What is the Chart length of a line representing an earth distance of 50 Kilometres on this chart? Give your answer in both Centimetres and Inches.**

4. **On a chart 5 Centimetres represents 7 Nautical Miles**
   a. Give the Scale of this chart as a Representative Fraction

b. Determine the distance in Inches on this chart which would represent the distance
flown by an aircraft in 5 minutes at a Groundspeed of 156 Knots.

5. **On a chart 14.8 Centimetres represents 20 Nautical Miles on the ground**
   a. Give the Chart Scale as a Representative Fraction
   b. Determine the length in Inches on this chart which would represent the distance flown
   by an aircraft in 9 minutes at a Groundspeed of 185 Knots.

6. **On a chart, One Centimetre represents 3.5 Kilometres**
   a. Give the Scale as a Representative Fraction
   b. What is the length in Inches on the chart which would represent the distance flown in
   4 minutes at a Groundspeed of 204 kilometres per hour.

Answers

1. a. 1:500,000
b. 1:1,000,000
c. 1:600,000 (approx)
d. 1:788,000
e. 1:116740
2. a. (1) 6 Centimetres (2) 2.36 Inches b. (1) 4.38 Inches (2) 11.12 Centimetres
3. a. 5 cm = 1.97inch b. Smaller
4. a. 1:260,000 b. 3.65 inches
5. a. 1:250,000 b. 8.1 inches
6. a. 1:350,000 b. 1.53 inches

Earth Magnetism

- At the center of earth there is a mass of molten liquid. This has the effect of making the Earth act magnetically as though there were a huge bar magnet running through it. Therefore, **earth itself acts as a huge magnet** with magnetic field lines running from N & S True Poles.

- **Heading**-
  - Is the direction in which the fore and aft axis of the aircraft is pointing; it may be measured from True, Magnetic, or Compass North.
  - Always measured from North
  - Always measured clockwise.

- **Bearing**
  - Is the direction of one point from another.
  - Always measured from North and clockwise.
  - Increases in Clockwise Direction and Decreases in an Anti-clockwise Direction.

- **Relative Bearing** – Measured from fore & aft axis of the aircraft.

- **True North –** It is measured with respect to geographical (true) north.

- **Magnetic North–**
  - As earth itself behaves as a magnet, the position of poles of that bar magnet do not coincide with the geographic north & south. This magnet is slightly tilted.
  - Due to this magnetic field, a freely suspended magnetic needle will point towards **Magnetic North**.

- **Compass North –**
  - Due to aircraft's magnetism, magnetic needle in an aircraft may not point towards True or Magnetic North. This indication given by compass is called Compass North.
  - It may or may not be same as Magnetic & True North.

- Directions measured from True North, Magnetic North and Compass North are called **True Direction, Magnetic Direction** & **Compass Direction** respectively.

- **Variation-**
  - Is the angle between True and Magnetic North.
  - Is measured in degrees East or West from True North.
  - Termed east if Magnetic North is East of True North.
  - Termed west if Magnetic North is West of True North.
  - Maximum Variation can be 180^0 E or W.

- **Isogonal**- It is the line on the surface of the Earth joining points of equal magnetic variation.

- **Agonic** is the name given to line joining places of zero magnetic variation.

- **Deviation** –
  - Is the angle measured between the direction indicated by a compass needle and the direction of Magnetic North.
  - It is termed east if Compass North lies to the East of Magnetic North.
  - It is termed west if Compass North lies to the West of Magnetic North.

Compass	Deviation	Magnetic	Variation	True
	5W	291	3E	
085	10E		8W	
	17W		11W	357
	20W	359	18E	
	2W		4E	350

- **Directive Force (H)**- It is the force which causes the alignment of the magnet with earth's field.

- **Magnetic Dip Angle -** The Earth Magnetic field is along the total line of force, shown as T. This can be resolved into a horizontal component (H) and a vertical component (Z).

  Due to Earth's Magnetic Field, Tan θ = Z/H
  Where θ = Angle of Dip.

- **Angle of Dip** is the angle of magnetic inclination OR the angle in the vertical plane between the horizontal and the Earth's magnetic field at a point.

- It varies from 0^0 at equator to 90^0 at poles.

- **Isoclinals** are lines on a map or chart joining places of equal magnetic dip.

- **Aclinic Lines** is the name given to isoclinals joining places of zero dip. Eg- Equator.

- **Note-** As the directive force (H component) reduces with increase in magnetic latitude, the magnetic compass system gives very poor indication at or near to the poles.

- Therefore, Magnetic compass is
  - **Most strong at or near equator and**
  - **Least strong at or near poles**.

Questions

1. **The sensitivity of a direct reading magnetic compass is:**
   a. Inversely proportional to the horizontal component of the earth's magnetic field.
   b. Proportional to the horizontal component of the earth's magnetic field.
   c. Inversely proportional to the vertical component of the earth's magnetic field.
   d. Inversely proportional to the vertical and horizontal components of the earth's magnetic field.

2. **What is the definition of magnetic variation?**
   a. The angle between the direction indicated by a compass and Magnetic North.
   b. The angle between True North and Compass North.
   c. The angle between Magnetic North and True North.
   d. The angle between Magnetic Heading and Magnetic North.

3. **At the magnetic equator:**
   a. Dip is zero
   b. Variation is zero
   c. Deviation is zero
   d. The isogonal is an agonic line

4. **Which of these is a correct statement about the Earth's magnetic field?**
   a. It acts as though there is a large blue magnetic pole in Northern Canada
   b. The angle of dip is the angle between the vertical and the total magnetic force.
   c. It may be temporary, transient, or permanent.
   d. It has no effect on aircraft deviation.

5. **Where is a compass most effective?**
   a. About midway between the earth's magnetic poles
   b. In the region of the magnetic South pole
   c. In the region of the magnetic North pole
   d. On the geographic equator

6. **The value of variation:**
   a. is zero at the magnetic equator
   b. has a maximum value of 180°
   c. has a maximum value of 45° E or 45° W
   d. cannot exceed 90°

7. **The agonic line:**
   a. is midway between the magnetic North and South poles
   b. follows the geographic equator
   c. is the shorter distance between the respective True and Magnetic North and South poles
   d. Follows separate paths out of the North polar regions, one currently running through Western Europe and the other through the USA.

8. The angle between True North and Magnetic north is known as:

a. deviation
b. variation
c. alignment error
d. dip

9. The value of magnetic variation on a chart changes with time. This is due to:

a. Movement of the magnetic poles, causing an increase
b. Increase in the magnetic field, causing an increase
c. Reduction in the magnetic field, causing a decrease
d. Movement of the magnetic poles, which can cause either an increase or a decrease

10. Isogonal lines converge as follows:

a. At the North Magnetic Pole
b. At the North and South Magnetic and Geographical Poles
c. At the North and South Magnetic Poles
d. At the Magnetic equator.

11. What is the maximum possible value of Dip Angle?

a. 66° b. 180° c. 90°
d. 45°

12. What is the dip angle at the South Magnetic Pole?

a. 0° b. 90° c. 180°
d. 64°

13. What is a line of equal magnetic variation?

a. An isocline
b. An isogonal
c. An isogriv
d. An isovar

14. If variation is West; then:

a. True North is West of Magnetic North
b. Compass North is West of Magnetic North
c. True North is East of Magnetic North
d. Magnetic North is West of Compass North

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
B	C	A	A	A	B	D	B	D	B	C	B	B	C

Dead Reckoning Navigation

- **Heading** is defined as the direction in which the fore and aft axis of the aircraft is pointing.

- **Track** is the direction of the aircraft's path over the ground. It may be measured from True or
Magnetic North.

- **Course** is a rarely used term which means 'desired track'.

- **Track Made Good (TMG) -** It is the actual path followed on ground. It may differ from required path due factors like wind.

- **Wind Velocity (V)** is the wind speed and **Wind Direction (W)** is the direction from which wind is coming and not which wind is going.

- **Drift-**
  - The angular difference between Heading & TMG.
  - Termed starboard if TMG is starboard of HDG.
  - Termed port if TMG is port of HDG.

- **Track Error-**
  - The angular difference between required track & TMG.
  - TE is starboard if TMG is starboard of required track.
  - TE is port if TMG is port of required track.

HDG^0(T)	W/V	TRACK^0(T)	TAS	GS
273	230/40		150	
181	150/30		90	
054	350/28		89	
084	255/55		210	
141	280/35		190	
305	350/16		100	
187	270/60		110	
310	045/45		320	

HDG^0(T)	W/V	TRACK^0(T)	TAS	GS
209		219	150	134
270		266	180	202
223		224	206	246
069		069	138	124
111		115	120	114
228		245	460	505
175		168	121	114

HDG^0(T)	W/V	TRACK^0(T)	TAS	GS
	040/40	155	140	
	280/27	226	94	
	320/14	198	136	
	190/52	284	260	
	270/83	132	544	
	310/105	262	572	
	300/60	355	620	

CAS	Pressure Altitude	SAT	TAS
140	10000	-10	
175	8000	+5	
220	15000	-22	
295	31000	-53	
160	12000	-10	
260	35000	-45	

Mach No.	SAT(^{0}C)	TAS
.82	-50	
.66		375
	-38	425
.77	-60	

.92		580
-47		415

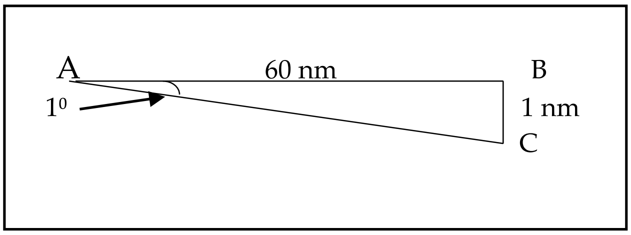
Indicated Altitude	SAT (^0C)	True Alt
10,000	-20	
18,000	-30	
6,000	+10	
27,000	-50	
21,500	-35	
8,000	+10	

Alt/ FL	Temp (^0C)	CAS (KT)	TAS (KT)	Track ^{0}(T)	W/V	HDG ^{0}(T)	VAR	Hdg ^{0} (M)	DEV	HDG (C)	GS (kt)	Dist (NM)	Time (min)
2000	-10	160		100	330/17		5W		1W			50	
4700	-2	170		227	300/10		3E		2E			112	
4000	-10	175		134	262/17		3W		1E			30	
2000	+2	170		157	110/12		6W		2W			38.5	
6000	-3	154		100	320/18		4E		1W			60	
5000	-15	150		050	110/45		8W		2E			13	
10000	-8	173		320	181/27		3E		-3			44	

Tr	W/v	Hdg T	Var	Hdg M	Dev	Hdg C	CAS	PA/Temp	Temp	TAS	GS	Dist	Time
315	045/55		7W		2E		190	3000	+15			55	
324	335/21		6E		1W		205	9000	-10			105	
071	030/55		11W		1W		142	14000	+5			50	
300	190/25		5E		3E		132	4000	-10			332	
019	300/25		9W		2E		130	5500	-17			110	
342	130/40		3E		1W		160	7500	0			92	
262	010/35		34E		2W		180	9500	0			145	
021	265/32		16W		4W		159	7000	+15			112	
033	310/50		7W		2E		202	22500	-30			46	
162	210/20		14W		3E		194	28000	-55			1312	
228	280/20		10W		2E		148	2000	+4			84	
351	060/30		12W		2W		123	3000	+16			95	
043	123/46		9E		1W		166	5000	-5			62	
032	120/30		4E		1E		123	2500	0			49	
162	140/40		9W		1W		96	3500	+5			130	
343	220/50		11E		1W		111	4000	-5			82	

1 in 60 Rule -

- The theory of One in 60 rule is based on the fact that a one degree of arc on the equator equals approximately 60 nm or 1′ = 1nm.



- 1 in 60 rule states that drift angle (θ) = $\dfrac{\text{Distance off track}}{\text{Distance covered/ Distance to go}} \times 60$
- This rule is applicable for $\theta \leq 20^{0}$ and is used to find off track diatances in flight.

Total Correction = Drift + Closing Angle.

APPLICATIONS OF THE 1 in 60 RULE -

1. Calculation of height on a glide slope-
   Height = G.S (TE) x distance to touchdown x 100.
2. Calculation of rate of descent to maintain a glide slope.-
   ROD (in feet/min) = 5 x Ground speed (kt) (for a 3° glideslope only)

3. Calculation of shallow slopes, particularly runway slopes-
   Change in ROD = 5 x change in speed (3° glideslopes only)

4. Calculation of distance off required track using radio aids such as VOR/DME-
   Angle off (TE) = $\dfrac{\text{Distance off (DO)}}{\text{DME range (DG)}} \times 60$

Questions

1. **You are flying from A to B. You find that your position is 60 nm outbound from A and 7 nm left of the required track. What is your track error angle?**

2. **You are flying from J to K, which is a required track of 045^{0}T. You find that your position is 80 nm outbound from J and 4 nm left of the required track. What is your TMG?**

3. **You are flying from L to M, which is a required track of 220^{0}T. You find that your position is 45 nm outbound from L and 3 nm right of the required track. What is your TMG?**

4. You are flying an instrument approach to an airfield and the required glide slope angle is 3^0. What height should you be passing when you are exactly 2 nautical miles from the touchdown point?

5. You are flying from S to T, which is a required track of 272^{0}T. You find that your position is 50 nm from T and 5 nm right of the required track. What track must you fly to arrive overhead T?

6. An aircraft leaves A to fly to B, 95 nm distance. Having flown 35nms, the aircraft position is found from a point is 7 nm right of track.
a. What is the track error?
b. What alteration of heading is required to fly direct to B?

7. An aircraft is flying due South. At 1000 hrs, point P bears 267°(T) from the aircraft. At 1006 hrs, point P bears 275° (T) from the aircraft. If the aircraft has a ground speed of 120kts, estimate the range of the aircraft from point P.

8. You are approaching runway on a glide slope of 3.5°.
   a) What height should you be at 2 miles range?
   b) What will be your ROD if ground speed is 120 kts?

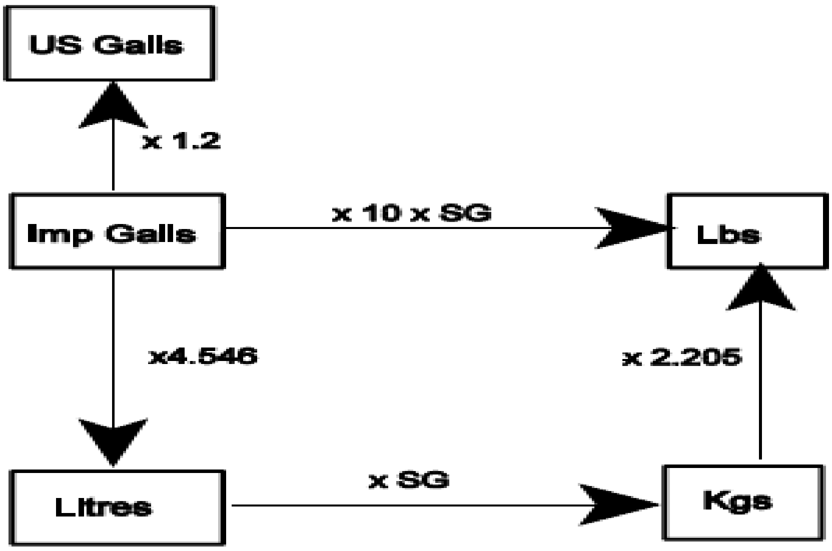
9. On the approach to London Heathrow runway 27, glide slope 3°, you reduce speed from 150 kts to 120 kts. What change should you make to your ROD to maintain glideslope?

10. You are flying an airway with a centreline QDM of 137 ° M towards VOR/DME 'A'. Your RMI reads 141° M/DME 90 nms.
   a. Are you left or right of centreline?
   b. What is your distance off the airway centreline?

Answers

1	7 L
2	42 T
3	224 T
4	600ft
5	226 T
6	a. 12° right b. 19° left (TE = 12°, CA = 7°)
7	90 nm
8	a.600ft b. 700ft/min
9	Decrease Rod by 150ft/min
10	a.Left b. 6nms

Conversion of unit –



Convergency

- Meridians are half GC joining the poles. They are parallel at equator and converge at the poles.
- At poles, they converge at an **angle equivalent to their ChLong**.
- **Convergency** is the angle of inclination between two meridians at a given latitude**.**
- If the latitude is higher, inclination between the meridians will be more. Hence, Convergency will be more. Therefore, Convergency is a **function of sin latitude.**

Convergency = ChLong x Sin Mean Latitude

- In NH, GC is always north of RL.
- In SH, GC is always south of RL.
- Higher the latitude, more the convergency.
- More the ChLong, more is the convergency.
- **Conversion Angle (C.A)** - Is the angular difference between GC and a RL bearing.
- **C.A = ½ Convergency =** ½ ChLong x Sin Mean Latitude.
- **C**.A is the function of Convergency, and is maximum at the poles.
- Requirement of C.A-
  - A/c follows RL track & Radio signals follow GC bearing.
  - On Earth GC is a straight line & RL is a curved line.
  - On Mercator chart, RL is a straight line & GC is curved line convex to nearer pole.
  - By applying C.A, we can convert GC bearing into RL bearing to plot radio bearings on Mercator charts.

Questions

1. **The convergency of the meridians through M and N which are in the southern hemisphere is 12°. If the RL track from M to N is 249°(T), what is the great circle track:**
   a. from M to N? **b. from N to M?**

2. **The rhumb line from position D (30°00′N 179°00′W) to position C is 090°(T).The great circle initial track from C to D is 287°(T). What is:**
   a. The great circle track from D to C?

b. The approximate latitude and longitude of position C ?

3. **Determine the value of convergency between J (5812N 00400W) and K (5812N 00600E). What is the rhumb line track from J to K? What is the initial great circle track from K to J?**

4. **The great circle track from A to B measures 227°(T) at A and 225°(T) at B. What is the convergency of the meridians through A and B and in which hemisphere are they?**

5. **The following waypoints are entered into an inertial navigation system (INS). WPT 1: 53°08′N 030W WPT 2: 53°08′N 020W WPT 3: 53°08′N 010W. The inertial navigation is connected to the automatic pilot on the route WP1 – WP2– WP3. The track change on passing WP 2 will be approximately:**
   a. A 8° increase b. A 4° decrease c. zero d. A 8° decrease

6. **Given that: A is N55° E/W 000° & B is N54° E 010°, if the initial true great circle track from A to B is 100° (T), what is the true Rhumb Line track at A?**
   a. 096°(T) b. 107°(T) c. 104°(T) d. 100°(T)

7. **The initial great circle track from B to A is 245°(T) and the RL track from A to B is 060°(T). If the mean latitude between A and B is 53° and the longitude of B is 02°15′E, what is the longitude of A?**

8. **A and B are both in the southern hemisphere and the convergency of their meridians is 8°. The initial great circle track from A to B is 094°(T). If the position of B is 23°00′S 20°00′W, what is the position of A?**

Answers

1	243°(T) & 075°(T)
2	073°(T) & 30°N 111°W
3	8.5°,090°(T) On same Latitude so a Rhumb Line Track, 274.25°
4	2^0 in NH
5	D
6	C
7	010° 15′ W
8	23° 00S 040° 30′ W

Projections

- The technique involved using a light source within the globe to project the latitude/longitude graticule on to a flat sheet of paper (Map or Chart) is called **Projection.**

- **Map** - All the geographical features are shown on map depending on its scale.

- **Chart**- They contain limited information for which the chart has been made. Eg- Jeppesen Approach & En-route Charts.

- Charts are divided into two categories-
  - Charts produced directly from a projection are called **Perspective** or Geometric Projections.
  - Charts produced by mathematical methods are called **Non-Perspective.**

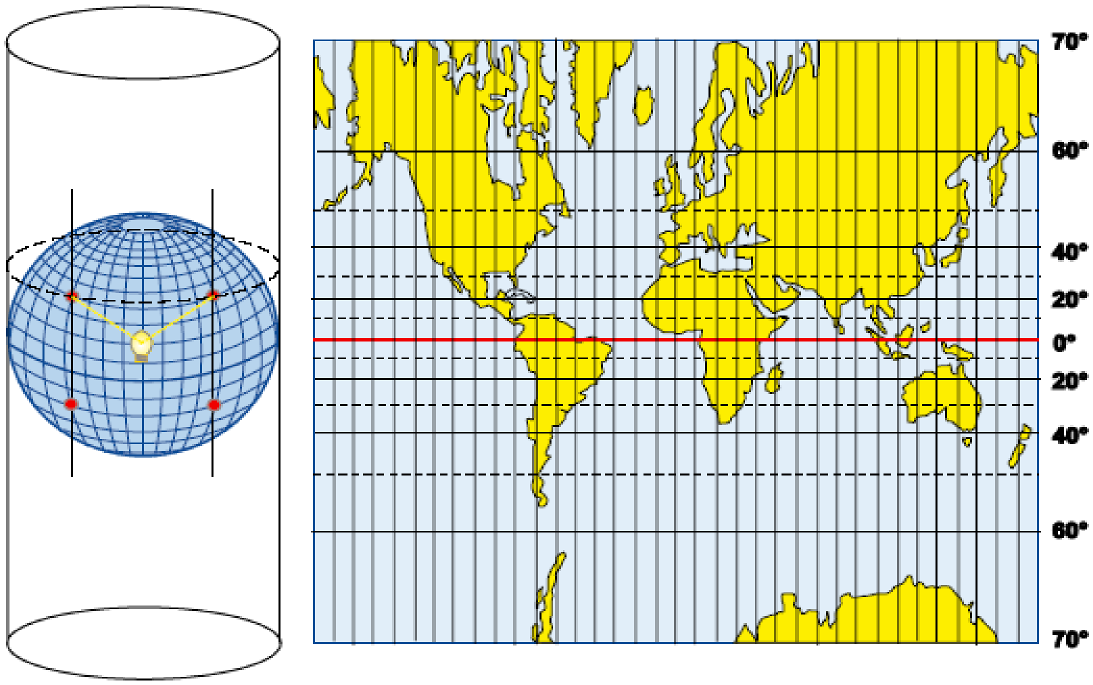
- **Reduced Earth (RE)** – It means the scale model of the earth on which the projection of the chart is based.

Properties of an ideal chart -

- ➢ Angles on the Earth's surface should be represented by the same angles on the chart.
- ➢ Scale should be constant and 'correct'.
- ➢ Areas should be represented with their true shape on the chart.
- ➢ Equal areas on the Earth's surface should be shown as equal areas on the chart.
- ➢ Rhumb Lines should be straight lines.
- ➢ Great Circles should be straight lines.
- ➢ Latitudes/Longitudes should be easy to plot
- ➢ Adjacent sheets should fit correctly.
- ➢ Coverage should be worldwide.
- ➢ Orthomorphic or Conformal Charts- Navigation bearings must be correctly represented and the parallels and meridians must cut each other at right angles. Also, scale must be same in all directions.

- **Types of Projection -**
  1. Azimuthal/Plane
  2. Cylindrical
  3. Conical

Direct Mercator Projection –



Properties of Direct Mercator	
Origin of Projection	Cylindrical The projection touches reduced earth at the equator. Projection is from the center of the sphere.
Scale	Correct on the equator. Elsewhere increases as the secant of latitude.
Graticule	Meridians are straight lines, equally spaced. Parallels of Latitude are unequally spaced parallel straight lines, with the spacing increasing away from the equator.
Chart Convergence	Correct at the Equator. At all other latitudes, chart convergence is less than earth convergence.
Rhumb Line	Straight Line
Great Circle	Curves convex to the nearer pole and concave to the equator. Equator and meridians are straight line.

Mercator scale expands as the secant of the latitude. This arises out of the departure formula. You will remember that:

Departure = change of longitude (min) × cos latitude.

Change of longitude = Departure/cos latitude

Change of longitude = Departure x secant latitude

1. If the scale of a Mercator chart at the Equator is 1:1,000,000, what is the scale at 60N (or S)?

a. 1:2,000,000
b. 1:1,000,000
c. 1:866,000
d. 1:500,000

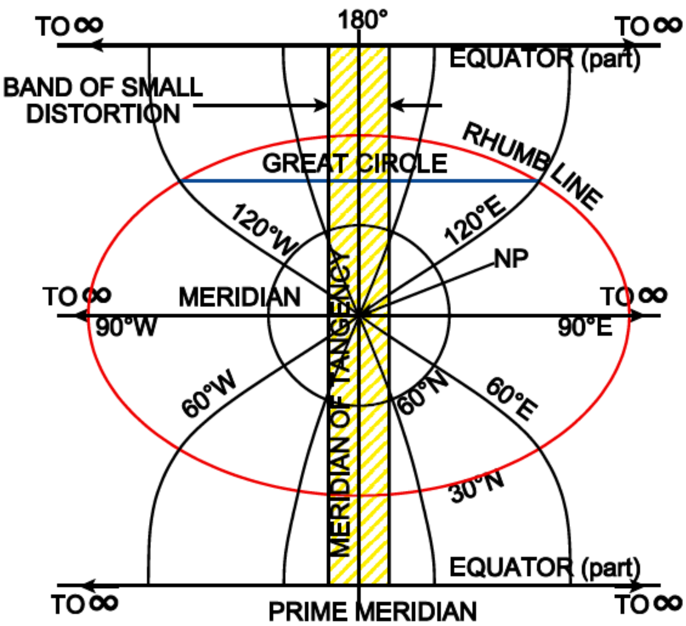
2. If the scale of a Mercator chart at 52S is 1:2,000,000, what is the scale at the Equator?

a. 1:3,250,000
b. 1:1,000,000
c. 1:866,000
d. 1:500,000

3. On a Mercator chart, the scale at 54S is 1:2,000,000. What is the scale at 25N?

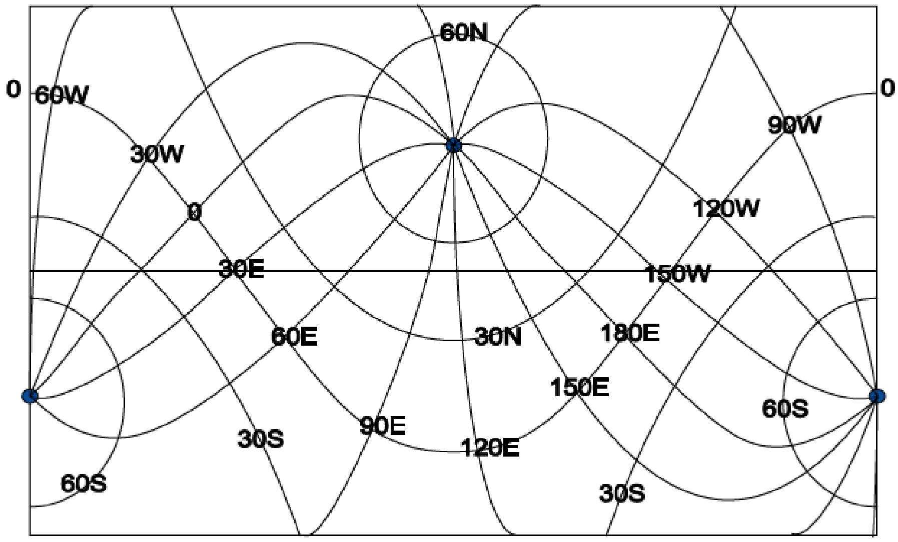
a. 1:2,000,000
b. 1:3,084,000
c. 1:1,121,000
d. 1:3,825,000

Transverse Mercator Projection-



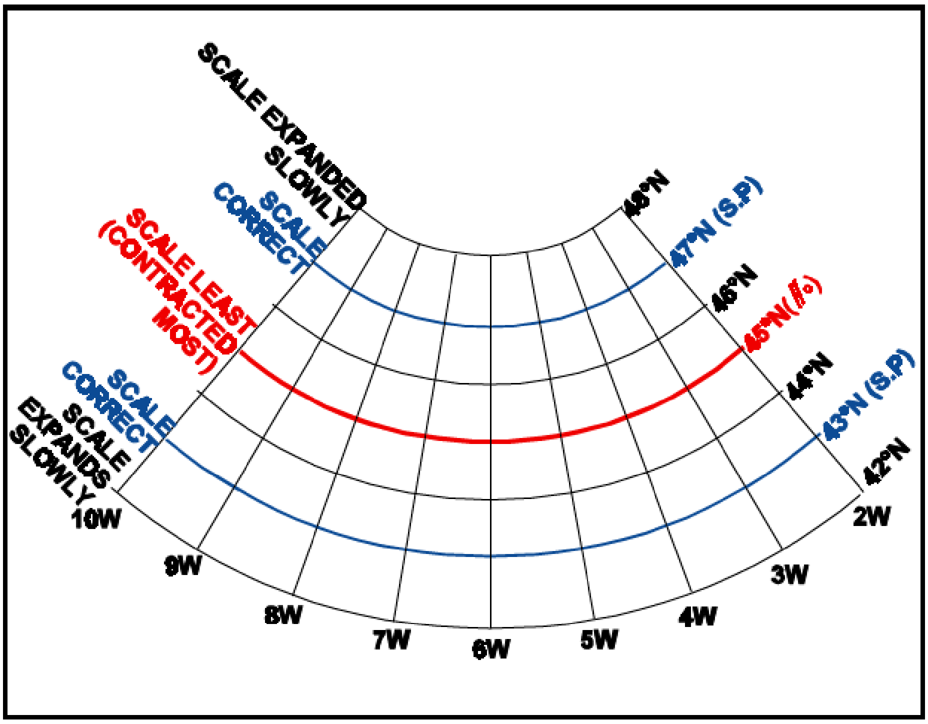
Properties of Transverse Mercator	
Origin of Projection	Cylindrical The cylinder the touches reduced earth at the selected meridian. Projection is from the center of the sphere.
Scale	Correct at the datum meridian. Expands away from the datum meridian as secant of great circle distance from the datum meridian.
Graticule	The Datum Meridian, The Equator, & Meridians 90^0 to the datum meridian are straight lines. Other Meridians are complex curve.
Chart Convergence	Correct at the Equator & Poles
Rhumb Line	Complex Curves
Great Circle	Complex Curves

Oblique Mercator Projection-



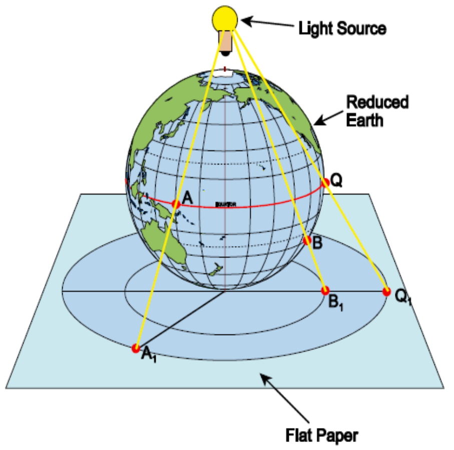
Properties of Oblique Mercator	
Origin of Projection	Cylindrical The cylinder touches the reduced earth along Great Circle of Tangency.
Scale	Correct at the Great Circle of Tangency. Expands as secant of great circle distance from the Great Circle of Tangency. Within 500nm of the GC of tangency may be used as a constant Scale chart.
Graticule	The Datum Meridian, The Equator, & Meridians 90^0 to the datum meridian are straight lines. Other Meridians are complex curve.
Chart Convergence	Correct along Great Circle of Tangency, Poles & Equator.
Rhumb Line	Complex Curves
Great Circle	Complex Curves but may be taken as straight lines close to Great Circle of Tangency.

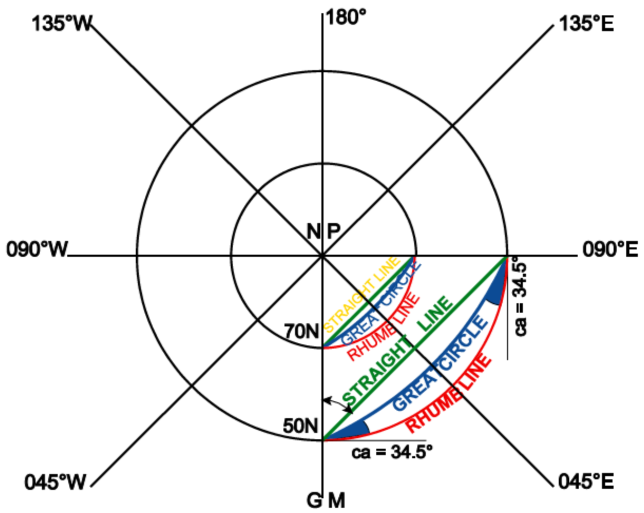
Lamberts Conical Projection-



Properties of Lambert Conical	
Origin of Projection	Conical The cylinder touches the reduced earth at parallel of tangency. Projection from the center of the sphere.
Scale	Correct at the Standard Parallels. Expands outside the standard parallels and contracts between the standard parallels and is at minimum at the parallel of origin.
Graticule	Meridians are straight lines converging towards pole of projection. Parallels of Latitude are arcs of circles, nearly equally spaced with their center at the pole of projection.
Chart Convergence	Correct at parallel of origin. Chart Convergence = ChLong x Sin parallel of origin.
Rhumb Line	Curves concave to the pole of projection. Meridians are straight lines.
Great Circle	Curves concave to parallel of origin. Are closest to a straight line at the parallel of origin.

Polar Stereographic Projection-





Properties of Polar Steographic	
Origin of Projection	Azimuthal The flat plate touches the reduced earth at the pole. Projection is from the opposite pole.
Scale	Correct at the Pole. Expands away from the pole as sec^2 ½ Co-Lat Scale is correct to within 1% to 78^0 N/S Scale is correct to within 3% to 70^0 N/S
Graticule	Meridians are straight lines radiating from pole. Parallels of Latitude circles centred on the pole. The spacing increases away from the pole. The Equator can be projected.

Chart Convergence	Correct at parallel of origin. Chart Convergence = ChLong x Sin parallel of origin.
Rhumb Line	Curves concave to the pole of projection. Meridians are straight lines.
Great Circle	Curves concave to parallel of origin. Are closest to a straight line at the parallel of origin.

Questions

1. **A direct mercator chart, meridians are:**
   a. Parallel, equally spaced, horizontal straight lines
   b. Converging curved lines
   c. Parallel, equally spaced, vertical straight lines
   d. Diverging curved lines

2. **On a normal Mercator chart, rhumb lines are represented as**:
   a. Curves concave to the Equator
   b. Curves convex to the Equator
   c. Complex curves
   d. Straight lines

3. **On a direct Mercator, Great Circles can be represented as:**
   a. Straight lines
   b. Curves
   c. Straight lines and curves

4. **On a direct Mercator, with the exception of the meridians and the Equator, Great Circles are represented as:**
   a. Curves concave to the Nearer Pole
   b. Curves convex to the Equator
   c. Curves concave to the Equator
   d. Straight lines

5. **The angle between a straight line on a Mercator chart and the corresponding GC is:**
   a. Zero
   b. Earth convergency
   c. Conversion Angle
   d. Chart Convergence

6. **At 60S on a Mercator chart, chart convergence is:**
   a. greater than Earth convergency
   b. "correct"
   c. less than Earth convergency
   d. equal to ch long x 0.866

7. **A Lamberts Conical conformal chart has standard parallels at 63N and 41N. What is the constant of the cone?**
   a. .891
   b. .788
   c. .656
   d. .707

8. **Scale on a Lambert's conformal conic chart**
   a. is constant
   b. is constant along a meridian of longitude
   c. varies slightly as a function of latitude and longitude
   d. is constant along a parallel of latitude

9. **Parallels of latitude except equator are:-**
   a. Rhumb lines
   b. Great circles
   c. curves convex to the nearer pole
   d. curves concave to the nearer pole

10. **Convergence on a Transverse Mercator chart is correct at:**
    a. the datum meridian only
    b. the datum meridian and the Equator
    c. the Equator and the Poles
    d. the Parallel of Origin

11. **Where is scale correct on a Transverse Mercator chart?**
    a. along the great Circle of Tangency
    b. at the Poles and the Equator
    c. along the Datum Meridian and at meridians at 900 to it
    d. at the Greenwich meridian

12. **What is the main use of a Transverse Mercator chart?**
    a. flying a specified Great Circle route
    b. flying an equatorial route
    c. mapping countries with a large N/S extent but a lesser E/W extent
    d. mapping countries with a large E/W extent but a lesser N/S extent

13. What is the main use of an Oblique Mercator chart?

a. flying a specified Great Circle route
b. flying an equatorial route
c. mapping countries with a large N/S extent but a lesser E/W extent
d. mapping countries with a large E/W extent but a lesser N/S extent

14. The angular difference, on a lambert conformal conic chart, between, the arrival and the departure track is equal to?

a. Departure Angle
b. Map Convergence
c. Secant of half latitude
d. Secant of longitude

15. On a Lambert Conformal Conic Chart earth convergency is most accurately represented at the?

a. Standard Parallels
b. Outside the Standard Parallels
c. At all points between the Standard Parallels
d. Parallel of Origin

16. On a Transverse Mercator Chart, Scale is exactly correct along the?

a. Median of Tangency
b. Prime Meridian
c. Standard Parallels
d. Latitude of Origin

17. On a Lambert Conformal Conic Chart the convergence at the meridian is?

a. The same as earth convergency at the parallel of origin.
b. The same as scale
c. Most accurate at the standard parallels
d. Least accurate at the standard parallels

Answers

1	2	3	4	5	6	7	8	9
C	D	C	C	C	C	B	D	A

10	11	12	13	14	15	16	17
C	A	C	A	B	D	A	A

Performance

Definitions of Terms used –

- **Atmosphere (ISA) – The atmosphere defined in the ICAO document.**
    a. Air is a perfect dry gas.
    b. The temperature at sea level is 15^0C.
    c. The pressure at sea level is 1013 hPa (29.92 in Hg).
    d. The temperature gradient from sea level to the altitude at which temp becomes -56.5^{0}C is 1.98^{0}C/ 1000ft.
    e. The density at sea level under ISA conditions is 1.225 kg/m^3 .
- **Alternate Airport** – An airport at which an aircraft may land if a landing at the intended airport becomes inadvisable.
- **Aircraft Classification Number (ACN)** - A value assigned to an airplane to show its load force. The aircraft classification number must be compared to the pavement classification number (PCN) of an aerodrome. The aircraft classification number may exceed the pavement classification number by as much as 50% but only if the maneuvering of the aeroplane is very carefully monitored otherwise significant damage may occur to both the aeroplane and the pavement.
- **Climb Gradient** – It is the ratio, in the same units, and expressed as a percentage of Change of Height.

  Climb Gradient (%) = Altitude (ft)/ Horizontal Distance(ft) x 100
- **Ceiling –**
    a. **Absolute Ceiling** means the pressure altitude at which the ROC is zero.
    b. **Service Ceiling** is a pressure altitude where ROC is a defined value.
- **Critical Engine**- The engine whose failure would most adversely affect the performance or handling qualities of an aircraft.
- **Contaminated runway**- A runway covered by more than 25% of water whose equivalent depth is 3mm.
- **Gross Performance** – The average performance that a fleet of aeroplanes should achieve if satisfactorily maintained and flown in accordance with techniques described.
- **Net Performance** – It is the Gross Performance diminished by a margin laid down by the appropriate authority.
- **Measured Performance** - Performance achieved by the manufacturer under test conditions for certification.

- **Unaccelerated Flight** - Flight at a steady speed with no longitudinal, lateral or normal acceleration.
- **Runway Slope** – The difference in height between two points on the aerodrome surface divided by the distance between those two points, in the same units, and expressed as a percentage.

Types of Altitudes

- **Indicated Altitude**- The reading on the barometer when it is set to the current barometric pressure.
- **Absolute Altitude** – The vertical distance of an airplane above the terrain (AGL).
- **Pressure Altitude** – The altitude indicated when the altimeter is set to 1013hPa or 29.92 mmhg
- **True Altitude** – The vertical distance of airplane above Mean Sea Level (MSL). It is used for most accurate calculation of altitude.

$$\mathbf{T.A = P.A + \frac{\{4 \times ISA\ Deviation \times PA\}}{1000}}$$

- **Density Altitude**
  - The height at which prevailing density occurs.
  - It is pressure altitude pressure altitude corrected for non-standard ISA temperature.
  - When pressure is standard, Pressure Altitude = Density Altitude.
  - Higher the temp, more is the DA and vice-versa.
  - Used primarily for A/c performance calculations.

D.A = PA + 118 (Act – ISA Temp in ^{0}C).

Types of Speeds

- V_A – Design Manoeuvering speed.
- V_{EF} – Engine Failure Speed (CAS at which critical engine is assumed to fail).
- V_{EF} - Max. Flap Extended Speed.
- V_{FTO} – Final Take-off Speed.
- V_{LE} – Max. Landing Gear Extended Speed.
- V_{LO} - Max. Landing Gear Operating Speed.
- V_{LOF} – Lift-Off Speed (Main wheel lift off from ground).
- V_{MBE} – Maximum Brake Energy Speed.
- V_{MC} – Min. control speed with critical engine inoperative.
- V_{MCA} - Min. control speed during T/O, Climb.
- V_{MCG} - Min. control speed on or near ground.
- V_{MCL} - Min. control speed during approach & landing.
- V_{MU} – Min. Unstick Speed.
- V_{NE} – Never Exceed Speed.
- V_{NO} – Normal Operating Speed.

- V_R – Rotation Speed.
- V_{REF}- Reference Landing Speed.
- V_S – Stall Speed or the minimum steady flight speed at which A/c is controllable.
- V_{SO} – Stall Speed or the minimum steady flight speed in the landing configuration.
- V_{S1}- Stall Speed or the minimum steady flight speed obtained in a specified configuration.
- V_{S1g}- The one-g stall speed at which the A/c can develop a lift force equal to its weight.
- V_X – Speed for best Angle of Climb.
- V_Y – Speed for best Rate of Climb.
- V_1 – Take-off Decision Speed. (Action Initiation Speed)
- V_2 – Take-off Safety Speed.
- V_{2min} - Min. Take-off Safety Speed.
- V_3 – Steady initial climb speed with all engines operating.

- **Air Minimum Control Speed** - The minimum speed at which directional control can be demonstrated when airborne with the critical engine inoperative and the remaining engines at take-off thrust. Full opposite rudder and not more than 5 degrees of bank away from the inoperative engine are permitted when establishing this speed. VMCA may not exceed 1.2 Vs.

- **Ground Minimum Control Speed**- The minimum speed at which the airplane can be demonstrated to be controlled on the ground using only the primary flight controls.

- **Equivalent Speed** – CAS corrected for adiabatic compressible flow for the particular altitude. In standard atmosphere at sea level, CAS = EAS.

- **Final Take-off speed** – Speed that exists at the end of the T/O path in the en-route configuration with one engine inoperative.

- **Flap Extended Speed** – Max speed permissible with wing flaps in a prescribed extended position.

- **Indicated Air Speed** – Speed shown on the pitot static airspeed indicator uncorrected for airspeed system errors.

- **Landing Gear Extended Speed** – Max speed at which an A/c can be safely flown with landing gear extended.

- **Landing Gear Operating Speed** – Max speed at which landing gear can be safely extended or retracted.

- **Reference Landing Speed** – The speed of the A/c, in landing configuration, at the point where it descends through the landing screen height in the determination of the landing distance for manual landings.
- **Take-off Safety Speed** – Reference airspeed obtained after lift-off at which the required one-engine-inoperative climb performance can be achieved.
- **True Airspeed** – Airspeed of an A/c relative to undisturbed air.
- **Hydroplaning Speed**- The speed at which the wheel is held off the runway by a depth of water and directional control through the wheel is impossible.
- **Minimum unstick speed**-The minimum speed demonstrated for each combination of weight, thrust, and configuration at which a safe takeoff has been demonstrated.
- **Engine Failure Speed** - The calibrated airspeed at which the critical engine is assumed to fail and is used for the purpose of performance calculations. It is never less than V_{MCG}.
- **Landing Minimum Control Speed**- The minimum speed with a wing engine inoperative where it is possible to decrease thrust to idle or increase thrust to maximum take-off without encountering dangerous flight characteristics.
- **Decision Speed** - It is the speed at which the pilot, in the event of a power unit failure, must decide whether to abandon or continue the take-off.

<u>Declared Distances</u>

- **Take-off Run Available (TORA) -** The length of the runway suitable for normal operations.
- **Take-off Distance Available (TODA)** - The length of the runway plus any clearway and cannot be more than one and one half times the TORA, whichever is the less.
- **Accelerate Stop Distance Available (ASDA)** - If a T/O is aborted the A/c can be brought to a stop on a runway or on a Stopway. It is equal to TORA plus Stopway.
- **Landing Distance Available -** The length of the runway suitable for landing operations.
- **Balanced Field** - A runway for which the Accelerate Stop Distance Available is equal to the Takeoff Distance Available.
- **Clearway** An area beyond the runway, not less than 152 m (500 ft) wide, centrally located about the extended centreline of the runway.
- **Stopway** An area beyond the take-off runway, no less wide than the runway and centered upon the extended centerline of the runway, able to support the aeroplane during an abortive take-off, without causing structural damage to the aeroplane.

PERFORMANCE CLASSIFICATION-

Category	Multi engine jet	Multi engine Turbo Prop	Piston Engine
Mass > 5700 kg Or Passenger Seats > 9	A	A	C
Mass ≤ 5700 kg Or Passenger Seats ≤ 9	A	B	B

Effect of Variable Factors on Take-off Distance –

1. **Mass -** Increased mass reduces acceleration and increases T/O distance in following ways:-
   - Acceleration - As mass increases, acceleration will decrease, which will increase the take-off distance.
   - The wheel drag - Because of the increased wheel friction, wheel drag will increase. Therefore, acceleration is reduced and the take-off distance will increase.
   - The take-off safety speed increases.

2. **Air Density** - Density is determined by pressure, temperature and humidity. Density affects:
   - The power or thrust of the engine.
   - The TAS for a given IAS.
   - The angle of the initial climb.

   Low Density gives reduced thrust and increases TAS for a given IAS, both effects increase the T/O distance.

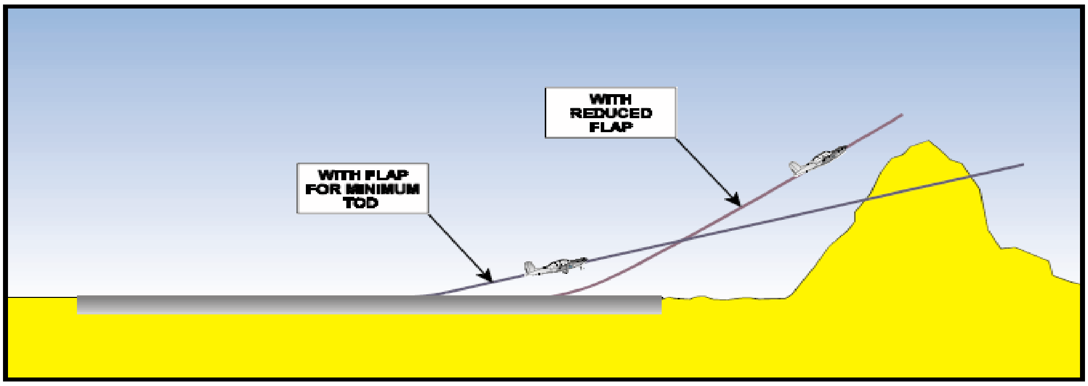
3. **Wind** - Headwinds will reduce the ground speed at the required take-off air speed and reduce the take-off distance. Tailwinds will increase the ground speed and increase the take-off distance. Note-
   a. The Regulations for all classes of aircraft require that in calculating the take-off distance, no more than 50% of the headwind component is assumed and no less than 150% of a tailwind component is assumed. This is to allow for variations in the reported winds during take-off.
   b. If the wind is a 90° crosswind the distance required to take off will be the same as the distance calculated for zero wind component.

4. **Runway Slope** - A downhill slope will increase the accelerating force, and reduce the T/O distance, whereas an uphill slope will reduce the accelerating force and increase the T/O distance.

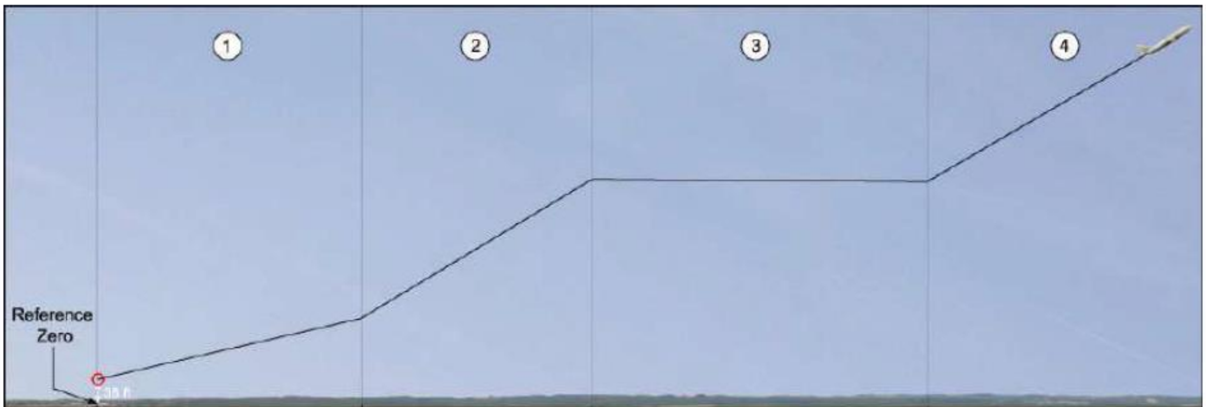
5. **Runway surface** - Even on a smooth runway there will be rolling resistance due to the bearing friction and tyre distortion. If the runway is contaminated by snow, slush or

standing water, there will be additional drag due to fluid resistance and impingement. Any contamination will increase the drag and hence increase the take-off distance.

6. **Flap Setting** - Flaps affect the CL max of the wing & the drag. Increasing flap angle increases CL max, which reduces stalling speed and take off speed. This reduces the take-off distance. The flap setting will also affect the climb gradient, and this will affect the Maximum Mass for Altitude and Temperature, which is determined by a climb gradient requirement, and the clearance of obstacles in the take-off flight path. Increasing the flap angle increases the drag, and so reduces the climb gradient for a given aircraft mass.



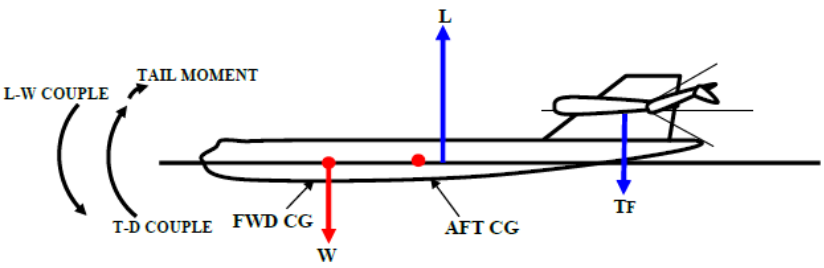
Take-off Segments



Starts	35 ft	Gear Up	> 400 ft AGL	Flaps Up V_{FTO} MCT
Actions	Select Gear Up	Climb to > 400 ft AGL	Retract Flaps Accelerate to V_{FTO} Set MCT	Climb to 1500 ft AGL
Speed	V_2	V_2	V_2 V_{ZF} V_{FTO}	V_{FTO}
Ends	Gear Up	> 400 ft AGL	Flaps Up V_{FTO} MCT	1500 ft AGL
Gradient	> 0 %	≥ 2.4 %	N/A	≥ 1.2 %
Gradient For 2 Engines	≥ 0.3 %	≥ 2.7 %	N/A	≥ 1.5 %
Gradient For 4 Engines	≥ 0.5 %	≥ 3.0 %	N/A	≥ 1.7 %

Center of Gravity and Datum

- **Centre of Gravity-** The point through which the force of gravity is said to act on a mass. The Centre of Gravity is also the point of balance and as such it affects the stability of the aircraft both on the ground and in the air.
- **Centre of Gravity limits-** The CG is not a fixed point it has a range of movement between a maximum forward position and a maximum rearward position which are set by the aircraft manufacturer and cannot be exceeded. The CG must be within the limit range at all times. They may also be given as a percentage of the mean chord of the wing. The mean chord is known as the Mean Aerodynamic Chord or MAC.



- **Datum** - A point along the longitudinal axis (center line) of the A/c designated by the manufacturer as the reference point from which all balance arms begin.
- **Balance Arm**- The distance from the aircraft's Datum to the CG position of a body of mass.

CG ON FWD LIMIT		CG ON AFT LIMIT	
STABILITY	↑	STABILITY	↓
STICK FORCES	↑	STICK FORCES	↓
MANOEUVRABILITY	↓	MANOEUVRABILITY	↑
DRAG	↑	DRAG	↓
V_S (STALLING SPEED)	↑	V_S	↓
V_R (ROTATION SPEED)	↑	V_R	↓
RANGE	↓	RANGE	↑
FUEL CONSUMPTION	↑	FUEL CONSUMPTION	↓
ABILITY TO ACHIEVE		**ABILITY TO ACHIEVE**	
1. CLIMB GRADIENT	↓	CLIMB GRADIENT	↑
2. GLIDE SLOPE	↓	GLIDE SLOPE	↑

Note –

- **Stalling speed will be highest for higher mass and FWD CG.**
- **Neutral Point is the point rearward side of the aft limit.**
- **Arm = Total Moment/ Total Weight.**

Performance Questions

1. **How is wind considered in the take-off performance data of the Aeroplane Operations Manuals?**
   a. Unfactored headwind and tailwind components are used.
   b. Not more than 80% headwind and not less than 125% tailwind.
   c. Since take-offs with tailwind are not permitted, only headwinds are considered.
   d. Not more than 50% of a headwind and not less than 150% of the tailwind.

2. **The take-off distance available is:**
   a. the total runway length, without clearway even if this one exists.
   b. the length of the take-off run available plus the length of the clearway available.
   c. the runway length minus stopway.
   d. the runway length plus half of the clearway.

3. **What will be the effect on an aeroplane's performance if aerodrome pressure altitude is decreased?**
   a. It will increase the take-off distance required.
   b. It will increase the take-off ground run.
   c. It will decrease the take-off distance required.
   d. It will increase the accelerate stop distance.

4. **An uphill slope:**
   a. increases the take-off distance more than the accelerate stop distance.
   b. decreases the accelerate stop distance only.
   c. decreases the take-off distance only.
   d. increases the allowed take-off mass.

5. **The result of a higher flap setting up to the optimum at take-off is:**
   a. a higher VR.
   b. a longer take-off run.
   c. a shorter ground roll.
   d. an increased acceleration.

6. **VR is the speed at which:**
   a. The aeroplane nose wheel is off the ground.
   b. The pilot initiate the action required to raise the nose wheel off the ground.
   c. The main wheels lift off the ground.
   d. The aeroplane rotates about the longitudinal axis.

7. **The stopway is:**
   a. at least as wide as the runway.
   b. no less than 152 wide.
   c. no less than 500 ft wide.
   d. as strong as the main runway

8. What is the effect of a contaminated runway on the take-off?

a. Increases the take-off distance and greatly increases the accelerate stop distance.
b. Increases the take-off distance and decreases the accelerate stop distance.
c. Decreases the take-off distance and increases the accelerate stop distance.
d. Decreases the take-off distance and greatly decreases the accelerate stop distance

9. A 'Balanced Field Length' is said to exist where:

a. The accelerate stop distance available is equal to the take-off distance available.
b. The clearway does not equal the stopway.
c. The accelerate stop distance is equal to the all engine take-off distance.
d. The one engine out take-off distance is equal to the all engine take-off distance

10. In relation to runway strength, the ACN:

a. Must not exceed 90% of the PCN and then only if special procedures are followed
b. May exceed the PCN by up to 10% or 50% if special procedures are followed
c. May exceed the PCN by a factor of 2
d. Must equal the PCN

11. Take-off distance available is?

a. Take-off run available plus clearway
b. Take-off run minus the clearway, even if clearway exists
c. Always 1.5 times the TORA
d. 50% of the TORA.

12. The take-off distance available is?

a. The total runway length, without clearway even if this one exists.
b. The length of the take-off run available plus any length of clearway available, up to a maximum of 50% of TORA.
c. The runway length minus stopway.
d. The runway length plus half of the clearway

13. A propeller aeroplane with nine or less passenger seats and with a maximum take-off mass of 5,700 kg or less is described as:

a. Unclassified.
b. Class C.
c. Class B.
d. Class A.

14. The speed for best rate of climb is called?

a. V_O.
b. V_Y.
c. V_X.
d. V_2.

15. A higher outside air temperature:

a. does not have any noticeable effect on climb performance.
b. reduces the angle of climb but increases the rate of climb.
c. reduces the angle and the rate of climb.
d. increases the angle of climb but decreases the rate of climb.

16. How does the best angle of climb and best rate of climb vary with increasing altitude?

a. Both decrease.

b. Both increase.

c. Best angle of climb increases while best rate of climb decreases.

d. Best angle of climb decreases while best rate of climb increases.

17. Which of the following provides maximum obstacle clearance during climb?

a. 1.2Vs.

b. The speed for maximum rate of climb.

c. The speed, at which the flaps may be selected one position further UP.

d. The speed for maximum climb angle Vx.

18. The absolute ceiling:

a. is the altitude at which the best climb gradient attainable is 5%.

b. is the altitude at which the aeroplane reaches a maximum rate of climb of 100 ft/min.

c. is the altitude at which the rate of climb is theoretically zero.

d. can be reached only with minimum steady flight speed

19. As long as an aeroplane is in a positive climb:

a. VX is always below VY.

b. VX is sometimes below and sometimes above VY depending on altitude.

c. VX is always above VY.

d. VY is always above VMO.

20. The maximum rate of climb that can be maintained at the absolute ceiling is:

a. 0 ft/min

b. 125 ft/min

c. 500 ft/min

d. 100 ft/min

21. A head wind will:

a. increase the rate of climb.

b. shorten the time of climb.

c. increase the climb flight path angle.

d. increase the angle of climb

22. VX is:

a. the speed for best rate of climb.

b. the speed for best specific range.

c. the speed for best angle of flight path.

d. the speed for best angle of climb.

23. During a climb with all engines operating, the altitude where the rate of climb reduces to 100 ft/min is called:

a. Thrust ceiling.

b. Maximum transfer ceiling.

c. Service ceiling.
d. Absolute ceiling.

24. A decrease in atmospheric pressure has, the following consequences on take-off performance:
a. a reduced take-off distance and degraded initial climb performance.
b. an increased take-off distance and degraded initial climb performance.
c. a reduced take-off distance and improved initial climb performance.
d. an increased take-off distance and improved initial climb performance.

25. Changing the take-off flap setting from flap 15° to flap 5° will normally result in:
a. a longer take-off distance and a better climb.
b. a shorter take-off distance and an equal climb.
c. a better climb and an equal take-off distance.
d. a shorter take-off distance and a better climb.

26. The climb "gradient" is defined as the ratio of:
a. true airspeed to rate of climb.
b. rate of climb to true airspeed.
c. the increase of altitude to horizontal air distance expressed as a percentage.
d. the horizontal air distance over the increase of altitude expressed as a percentage.

27. The first segment of the take-off flight path ends:
a. at completion of gear retraction.
b. at completion of flap retraction.
c. at reaching V2.
d. at 35 ft above the runway

28. During take-off, the third segment begins:
a. when acceleration to flap retraction speed is started.
b. when landing gear is fully retracted.
c. when acceleration starts from VLOF to V_2.
d. when flap retraction is completed.

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14
D	B	C	A	C	B	A	A	A	B	A	B	C	B

15	16	17	18	19	20	21	22	23	24	25	26	27	28
C	A	D	C	A	A	C	D	C	B	A	C	A	A

Flight Planning Questions

1. **Distance A to B = 600 nms, climb 30 mins@ 160 k GS, cruise GS= 250 k descent 20 mins @ 180 kts GS, flight time is equal to -**
   a. 2:21
   b. 2:25
   c. 2:30
   d. 2:41

2. **During climb from 1000 feet PA to 9000 feet PA, 50 air nms are covered in 20 min, track 090[T] W/V 270/30, calculate ROC and distance covered.**

3. **Given -**
   Climb Fuel = 1500 kg, distance = 130 nms
   Descent Fuel = 500 kgs, distance 120 nms
   TOW = 55 tons, Landing weight = 49 tons
   Calculate mid cruise weight.

4. **An a/c climbs under following conditions -**
   Set course at = 4000 feet PA ROC = 500 fpm
   Level out at = 16000 feet PA CAS = 130 K
   Temp at set course PA = ISA + 15 and at 16000 Feet ISA-5.
   Find time and distance for climb under nil wind.

5. **An a/c descends to the airport under following conditions**
   Cruising altitude = 6500 ft airport elevation = 700 ft
   Descent to 800 ft agl ROD = 500 fpm
   Average TAS = 110 kts Track = 335^0
   Average W/V = 060/15 VAR/DEV = 3^{0}w/+1^0
   Average fuel consumption = 8.5 gph.
   Calculate time, Co[c] distance and fuel consumed during descent.
   a. 12 min, 348, 17 nms, 1.8 gals
   b. 10 min, 345, 18 nms, 1.4 gals
   c. 10 min, 346, 16 nm, 1.4 gals

6. **Given -**
   Trip fuel = 9000 lbs
   Alternate fuel = 3000 lbs
   Tolerance = 10% of trip fuel
   Holding ½hr = @2000 lbs/hr
   Taxi/circuit/ldg = 800 lbs
   Calculate total fuel required in imp. Gals [SG= 0.72]

7. **Given -**
Total fuel = 12000 lbs **Reserve = 25% of flight fuel**
Avg fuel consp.= 1500 lbs/hr **TAS = 180 kts**
Safe distance to fly under nil wind is?

8. **Given -**
Distance to go = 68 nms **PA/ temp = 7000ft/+10^{0}C**
Track = 175 [T] **W/V = 295/25**
To reach the destination in 25 min ac should maintain a TAS of?

9. **Track 220 [T] W/V = 220/30 TAS = 250 K, FUEL FLOW= 8200 KG/HR. FIND GROSS FUEL FLOW/NM.**
a. 37.27 kg
b. 32.7 kg
c. 290.4 kg
d. 327 kg

10. **Convert 1000 Liters of fuel [sg=0.8]**

IMP GALS	USG	WT IN KGS
a. 264	220	800
b. 220	264	800
c. 125	150	8000

11. **Climb time = 40 mins with fuel flow = 3000 lbs/hr, cruise 130 mins with fuel flow = 2000 lbs/hr, descent 30 mins fuel flow= 1000 lbs/hr. Trip Fuel?**
a. 7500 lbs
b. 4833 lbs
c. 4583 lbs
d 6833 lbs

12. **Descend from FL 150 to PA 1000 feet at CAS 180 k hwc 20 k, temp at mid latitude= ISA +20. ROD 500 fpm. Descent distance nms __ mins**
a. 88/20
b. 75/28
c. 88/28

ANSWERS

1. D	5. B	9. A
2. 60NM, 400 FPM	6. 2042	10. B
3. 51.5 T	7. 1152	11. D
4. 24 MIN, 61 NM	8. D	12. C